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Practice Test #11

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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

Ezra Pound's poetry can be hard to _____: it is dense, experimental, and so full of references and allusions that many readers have a difficult time even identifying the poems' subjects.

Which choice completes the text with the most logical and precise word or phrase?

- A) comprehend
- B) dislike
- C) interrupt
- D) overlook

2

The unique *subak* water management system used to irrigate the rice paddy fields of the Indonesian island of Bali has a rich cultural, philosophical, and historical significance dating back to the ninth century. The many elements of *subak*—terraces, canals, and water temples—are _____: they are joined together into a single cohesive unit.

Which choice completes the text with the most logical and precise word or phrase?

- A) outmoded
- B) informal
- C) interconnected
- D) optional

3

Although the government of the Soviet Union attempted to _____ Georgi Vladimov’s novel *Faithful Ruslan*, copies of the book circulated in secret among readers in several parts of the country.

Which choice completes the text with the most logical and precise word or phrase?

- A) replicate
- B) critique
- C) import
- D) suppress

4

Scholars long thought that the initial spread of silk beyond China occurred in the second century CE, but this view has been _____ by new archaeological evidence from South Asia that reveals that the people of the Indus Civilization made use of silk at least 1,000 years earlier.

Which choice completes the text with the most logical and precise word or phrase?

- A) investigated
- B) misinterpreted
- C) anticipated
- D) contradicted

5

A casual description of Scherezade García’s 2019 mural *Blame It on the Bean: The Power of Coffee* can make the work seem _____—a painting that is housed in a coffee shop and that depicts three women drinking coffee may not sound particularly ambitious—but in fact the work is a complex, dynamic meditation on gender and the legacy of colonialism that demands serious attention.

Which choice completes the text with the most logical and precise word or phrase?

- A) unassuming
- B) shrewd
- C) incongruous
- D) pretentious

6

The following text is adapted from Akwaeke Emezi’s 2019 novel *Pet*. Jam is a teenager who lives with her father, Aloe, and her mother, Bitter, who is a painter.

Bitter finished the painting in the dark morning of a day—it was well past midnight when Jam heard the studio door creak open. She stared into the velvet black of her room and listened to her mother’s footsteps walking in her [mother] and Aloe’s bedroom. There was a weight thrumming through the floorboards in a low song, and that was how Jam knew the painting was done. Bitter’s feet were singing the news.

©2019 by Akwaeke Emezi

Which choice best states the function of the underlined sentence in the text as a whole?

- A) It indicates that Jam is more interested in music than in art.
- B) It adds to the idea that Bitter’s footsteps reveal something to Jam.
- C) It indicates that Bitter always sings when working on a painting.
- D) It describes Aloe’s reaction upon seeing the painting for the first time.

7

The following text is from Bram Stoker’s 1911 novel *The Lair of the White Worm*. Adam is meeting his great-uncle Richard at a port.

The meeting so auspiciously begun proceeded well. Adam, seeing that the old man was interested in the novelty of the ship, suggested that he should stay the night on board, and that he would himself be ready to start at any hour and go anywhere that the other suggested. This affectionate willingness to fall in with his own plans quite won the old man’s heart. He warmly accepted the invitation, and at once they became not only on terms of affectionate relationship, but almost like old friends.

Which choice best states the main purpose of the text?

- A) It states the reasons why Adam and his great-uncle Richard decide to sleep on the ship rather than finding lodging on land.
- B) It showcases how Adam’s flexibility and consideration strengthen his relationship with his great-uncle Richard.
- C) It describes why Adam and his great-uncle Richard are excited for their upcoming journey on the ship.
- D) It contrasts great-uncle Richard’s wary first impressions of Adam with his ultimate affection toward him.

8

Curious about how people visually perceive objects in their dreams, Stephen LaBerge and team recruited lucid dreamers—people aware that they’re dreaming as it’s happening—for a research study. These participants were reliably able to signal when they had entered a dream state; the team then observed participants’ eye movements as they slept. The smoothness with which participants’ eyes tracked objects in their dreams closely matched how sighted people who are awake visually track objects around them, suggesting to the team that the brain perceives dream objects as the product of something other than pure imagination.

Which choice best states the function of the underlined portion in the text as a whole?

- A) To offer key evidence that undermines LaBerge and team’s initial hypothesis
- B) To show the unexpected result that led LaBerge and team to change the focus of their study
- C) To illustrate an important real-world implication of LaBerge and team’s main finding
- D) To identify a comparable circumstance that helps justify LaBerge and team’s conclusion

9

Text 1

Good art often challenges and disrupts social and aesthetic norms, but the creation of public art—paintings, sculptures, and performance pieces displayed in nonmuseum or nontheatrical public settings—typically requires broad agreement among artists, civic officials, and community members about the works’ message and artistic goals. Public art that fails to appease everyone by being sufficiently aesthetically and conceptually bland almost inevitably provokes backlash.

Text 2

Public art is commonly displayed in spaces intended for purposes other than meaningful aesthetic engagement. Some critics of public art therefore note that norm-defying pieces that aren’t effectively integrated within their surroundings in a manner that primes passersby to appreciate the pieces’ merits (as is often the case) tend to be regarded more unfavorably than similarly provocative art encountered in museums is.

Based on the texts, how would the critics mentioned in Text 2 most likely respond to the underlined claim in Text 1?

- A) By arguing that the reason members of the general public might disagree about a public artwork’s merits is unrelated to the unconventionality of its appearance and ideas
- B) By agreeing with the idea that only works of art that are universally appealing are suitable for displaying in public spaces
- C) By disputing the notion that civic leaders and community members are easily placated by art that is intended mainly to reinforce social norms
- D) By contending that the kinds of reactions controversial public artworks often receive aren’t exclusively the result of attributes inherent in the works themselves

10

The average age at which people in the United States start businesses is 35. Economist Andrés Hincapié studied why young adults are relatively less likely to start businesses and whether there are ways to increase entrepreneurship in early adulthood. Hincapié found that one impediment is lack of knowledge about the practical details of how businesses are started; he further found that simply providing young adults with good informational resources on the topic significantly alleviates this problem.

Based on the text, what would Hincapié most likely say is a promising way to increase entrepreneurship in early adulthood?

- A) Creating social networks of young adults who are interested in starting a business
- B) Encouraging young adults to brainstorm business ideas
- C) Providing young adults with practical information about how to start a business
- D) Giving young adults training opportunities at a variety of businesses

11

Few animals are known to spit: among them are humans, cobras, and camels. But in January 2022 at a nature preserve in southern England, bird-watcher Clare Jacobs observed a gray seal spitting a jet of water at a white-tailed eagle flying overhead. Seals had never been seen spitting before. Biologist Sean Twiss, who studies gray seals, believes that the seal may have been attempting to scare the eagle away from a food source or that the seal may have just been playing.

Which choice best states the main topic of the text?

- A) Bird-watching in southern England
- B) A previously unseen behavior of gray seals
- C) How white-tailed eagles defend their territory against other predators
- D) Differences between gray seals and white-tailed eagles

12

The following text is from Mark Haber's 2022 novel *Saint Sebastian's Abyss*. The narrator and Schmidt are both art critics.

When my first wife admitted to Schmidt over dinner that she didn't find art, painting in particular, especially compelling, Schmidt winced, set down his fork, and sighed dramatically; he then excused himself, explaining an appointment he'd forgotten about had suddenly and inexplicably been remembered, while making it abundantly clear there was no appointment at all.

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Based on the text, what is notable about Schmidt's behavior?

- A) Schmidt is only given to theatrical behavior when in the company of the narrator and his first wife.
- B) Schmidt's absentmindedness regarding his schedule is uncharacteristic of him.
- C) Schmidt's departure is occasioned by the resumption of a previous disagreement with the narrator's first wife about a particular painting.
- D) Schmidt conveys his feelings about one of his dining companions without explicitly stating them.

13

The following text is from Mick Herron's 2023 novel *The Secret Hours*. The narrator is describing members of a government committee.

Finally, and adding much-needed gravitas, was Sir Winston Day, whose features seemed moulded to adorn a bust, or possibly a stamp, and whose forehead was so evidently bulging with grey matter that it would have been impertinent to inquire too closely into the actual achievements his half century of public service had produced. His recently published memoirs possibly cast light on this enigma, but given that such details were not provided until after the thirty-page mark, they might as well have remained state secrets.

©2023 by Mick Herron

Based on the text, which choice best describes Sir Winston Day?

- A) He has the appearance of a distinguished figure, but it is uncertain whether he has accomplished anything to earn distinction.
- B) He looks like a person worthy of respect, but his memoirs reveal that some of his actions were dishonorable.
- C) He would be a celebrated public figure if his achievements did not have to be kept secret.
- D) He has maintained a modest profile even though he has served the public capably for many years.

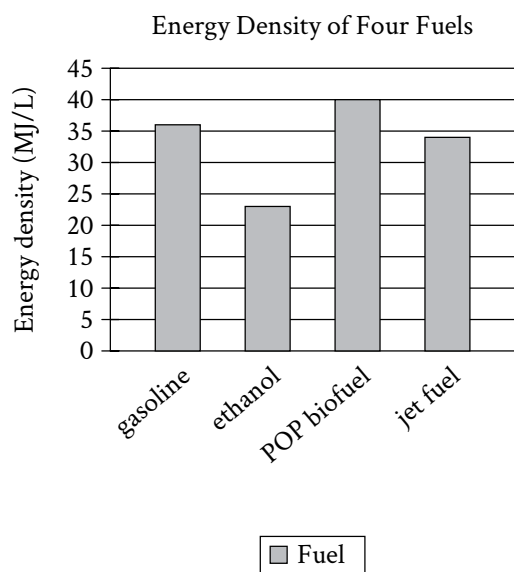
14

Studies of ocean wave breaking have predominantly focused on traveling waves (those propagating along the horizontal plane), so Mark McAllister et al. utilized a circular wave tank to produce and study spike waves, axisymmetric standing waves that can erupt vertically when traveling waves propagating in opposing directions intersect. Traveling waves break when wave steepness (height-to-length ratio) passes a critical threshold; breaking thus constrains wave height. McAllister et al. found that spike waves can exceed that constraint, as other factors than just steepness (e.g., jet stability and cavity shape) mediate spike-wave breaking.

Which choice best states the main idea of the text?

- A) Previous studies have suggested that steepness mediates breaking in traveling waves, but the study by McAllister et al. shows that jet stability and cavity shape may also influence breaking in such waves.
- B) The process of breaking limits the height of traveling waves, but the study by McAllister et al. suggests that spike waves can exceed those limits if their height-to-length ratio reaches a critical threshold.
- C) McAllister et al. suggest that spike waves can form when traveling waves propagating in opposing directions intersect and that spike waves tend to be higher than traveling waves.
- D) The study by McAllister et al. suggests that when traveling waves intersect in specific ways, the resulting wave may be higher than would be expected based on the properties of traveling waves.

15



A team of researchers used bacteria to create a biofuel (a renewable fuel produced from plants, algae, or other living materials). The team called the new fuel POP biofuel. To determine how much energy is stored in POP biofuel compared to other fuels, the team calculated the fuels' energy density, in megajoules per liter (MJ/L). The team found that the fuel with the highest energy density is _____

Which choice most effectively uses data from the graph to complete the sentence?

- A) jet fuel.
- B) POP biofuel.
- C) ethanol.
- D) gasoline.

16

Founded in 1965 and originally established as a cultural extension of the United Farm Workers—a union representing many Mexican American agricultural workers at the time—the theater troupe El Teatro Campesino has achieved recognition as a source of inspiration for subsequent Chicano theater companies and as a contributor to the dramatic arts. In an article about the company, a theater historian posits that a significant stylistic influence on El Teatro's early performances was the audience-mediated slapstick comedy of *carpa* theater, vaudeville-style shows popular in Mexico and the US Southwest in the 1920s and '30s.

Which quotation from the article would best illustrate the theater historian's claim?

- A) "The company presented *actos*, short comedy sketches, that often relied on exaggerated physical humor to groups of agricultural workers, whose reactions—enthusiastic cheers of appreciation and, occasionally, loud boos of disapproval—promoted improvisation."
- B) "The company was focused on the reality of the present situation and discovered that humor was often found in that reality; consequently, comedy became a tool to convey social critique while entertaining and inspiring audiences."
- C) "The company relied heavily on satire, humor, and references to contemporary popular culture as well as a make-do aesthetic—often referred to as *rasquache*—that reflected not only the troupe's limited financial resources but also its sociopolitical message."
- D) "The members of the company, which in addition to founder Luis Valdez consisted entirely of nonprofessional actors, traveled into farm fields, where they, with minimal props and costumes, performed comedy in the form of brief humorous vignettes."

17

Researchers Eugeni Vidal-Tortosa and Robin Lovelace looked at the relationship between street lighting in a city and people's willingness to ride a bicycle. Their results suggest that poor street lighting can deter new or inexperienced cyclists from riding in a city but has little effect on experienced cyclists. Therefore, increasing the number of streetlights in a city could potentially _____

Which choice most logically completes the text?

- A) decrease the number of new or inexperienced cyclists riding in the city.
- B) increase the number of experienced cyclists riding in the city.
- C) decrease the number of experienced cyclists riding in the city.
- D) increase the number of new or inexperienced cyclists riding in the city.

18

Like many other genera of wild bees, bumblebees have in recent decades experienced population collapse caused by, among other factors, habitat destruction and climate variation. Bumblebees are also one of the most researched bee genera, second only to honeybees. As a result, ecologists have gained much of their insight about wild-bee declines from bumblebees. In a 2021 paper, zoologist Guillaume Ghisbain notes that bumblebees are among the relatively few wild-bee genera that display social behaviors and dietary generalism (ability to obtain nectar and pollen from a diversity of plant species), two traits that are associated with increased resilience to some specific environmental changes. Ghisbain therefore contends that _____

Which choice most logically completes the text?

- A) although bumblebees and many other wild bees have experienced similar population declines in the past, compared with other wild bees, bumblebees are likely at greater risk of being harmed by climate variation than by habitat destruction.
- B) although bumblebees have been more extensively studied than most wild bees, researchers should not use bumblebees to draw conclusions about the decline of other wild bees, even ones with feeding patterns and levels of sociability that are similar to those of bumblebees.
- C) because bumblebees and other bees with generalist diets are less negatively affected by environmental stress than bees with specialized diets are, they are less likely to experience major population changes in the future than bees with specialized diets are.
- D) because the responses of bumblebees and other wild bees to environmental threats are not always comparable, researchers need to exercise caution when extrapolating information about wild-bee population declines from bumblebees.

19

When using a search engine, many people click just the first one or two results. Click restraint is the practice of scanning a search results page and evaluating what you see before deciding which link _____

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) should you choose.
- B) you should choose?
- C) should you choose?
- D) you should choose.

20

Following her debut album release in 2002, Mexican singer-songwriter Natalia Lafourcade quickly shot to fame. By 2023, she _____ one of the most celebrated musicians in Latin America, having released twelve albums and won seventeen Latin Grammy awards—more than any other female artist in history.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) will become
- B) becomes
- C) will have become
- D) had become

21

The rough frog is a species of amphibian native to Australia. Currently, the frog's range _____ parts of northern New South Wales and southeastern Queensland.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) includes
- B) included
- C) will have included
- D) had included

22

According to the original text of the US Constitution, written in 1787, the presidential candidate receiving the second-most Electoral College votes becomes vice president. The 12th amendment, ratified in _____ separated the elections for the two offices.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) 1804—
- B) 1804,
- C) 1804:
- D) 1804

23

The musical scores of Japanese composer Hiroyuki Sawano are famous for their mysterious titles. Laden with emojis and seemingly meaningless words, and driven largely by Sawano’s “personal feeling and mood,” _____

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) the listener can approach each piece free from expectations because of the titles.
- B) the titles allow the listener to approach each piece free from expectations.
- C) each piece can be approached by the listener free from expectations because of the titles.
- D) the listener’s approach to each piece because of the titles can be free from expectations.

24

Artist Yto Barrada’s exhibition *Ways to Baffle the Wind* incorporates sculptures, textiles, and films. Barrada’s pieces, utilizing elements as disparate as plant-dyed fabrics, wire crab traps filled with stones, and cotton balls dangling above a fan, _____ the ways humans attempt to organize and regulate nature.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) explore
- B) has explored
- C) explores
- D) exploring

25

Helical _____ widely understood to confer stability and efficiency in the locomotion of a variety of microscopic organisms—including bacteria, eukaryotic algae, and ciliates—bestows similar advantages, albeit via different propulsive modes, to larger oceanic macroplanktons, such as salps.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) swimming, is
- B) swimming is
- C) swimming has been
- D) swimming,

26

The tiny transistors that control the flow of electricity in modern devices may one day be made of wood. Researchers in Switzerland have found a way to use heat and chemicals to widen the grooves in dry pieces of balsa wood. _____ these grooves become wide enough that electrical conductors can be passed through them.

Which choice completes the text with the most logical transition?

- A) For example,
- B) Previously,
- C) In contrast,
- D) As a result,

27

Al-Andalus, the historical region of the Iberian Peninsula that includes most of modern-day Spain, was ruled by various Arabic-speaking Muslim states between the eighth and fifteenth centuries. _____ many Arabic words, such as “alacrán”—meaning “scorpion”—made their way into the Spanish language.

Which choice completes the text with the most logical transition?

- A) For example,
- B) Instead,
- C) Specifically,
- D) Consequently,

28

Imagine a magazine that a reader has thrown away. This magazine is post-consumer waste, as it became waste after reaching the consumer. _____ the paper scraps left over from printing the magazine are pre-consumer waste, as they became waste before reaching the consumer.

Which choice completes the text with the most logical transition?

- A) By contrast,
- B) For example,
- C) As a result,
- D) Specifically,

29

In astrophysics, a ring of debris orbiting a larger object within the object’s Roche limit is expected to persist as a ring, whereas a ring of debris orbiting outside this limit would likely accrete into a satellite (e.g., a moon). Bruno Morgado and colleagues, _____ detected a dense ring of material orbiting the trans-Neptunian object Quaoar at a distance of 2,500 miles, well outside the calculated Roche limit of 1,100 miles, that has remained intact.

Which choice completes the text with the most logical transition?

- A) though,
- B) for example,
- C) fittingly,
- D) likewise,

30

While researching a topic, a student has taken the following notes:

- “Organ²/ASLSP (As Slow as Possible)” is a musical piece by avant-garde composer John Cage.
- A specially designed automated organ in St. Burchardi Church in Halberstadt, Germany, began playing the piece in 2001.
- It is scheduled to stop playing the piece in 2640.
- The performance will last 639 years.
- It will be the longest continuous musical performance in history.

The student wants to indicate how long John Cage’s musical piece will last. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) John Cage is the composer of the musical piece “Organ²/ASLSP (As Slow as Possible).”
- B) “Organ²/ASLSP (As Slow as Possible)” is a musical piece currently being played in St. Burchardi Church in Halberstadt, Germany.
- C) Lasting 639 years, John Cage’s musical piece will be the longest continuous musical performance in history.
- D) An organ in St. Burchardi Church in Halberstadt, Germany, began playing a musical piece by avant-garde composer John Cage.

31

While researching a topic, a student has taken the following notes:

- 1914: British explorer Ernest Shackleton and a small crew embarked on an expedition to Antarctica.
- 1915: Shackleton’s ship *Endurance* became stuck in ice before eventually breaking apart and sinking.
- 1916: After more harrowing sea-ice adventures, the entire crew was rescued.
- 1959: Historian Alfred Lansing wrote a book called *Endurance: Shackleton’s Incredible Voyage*.
- 2001: Filmmaker George Butler released a documentary called *The Endurance: Shackleton’s Legendary Antarctic Expedition*.
- 2022: The wreckage of *Endurance* was discovered at the bottom of Antarctica’s Weddell Sea.

The student wants to provide a historical overview of the Shackleton expedition. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Leaving in 1914 for Antarctica, Shackleton and his crew underwent many harrowing sea-ice adventures, including losing their ship in 1915, before being rescued in 1916.
- B) In 1914, the Shackleton expedition sailed to Antarctica, where, in 1916, they rescued the crew of a ship that had sunk, *Endurance* (the wreckage of which was discovered in 2022).
- C) Shackleton’s expedition has inspired a 1959 book, a 2001 film, and a 2022 discovery.
- D) Alfred Lansing wrote about the history of Shackleton’s 1914–16 expedition in the book *Endurance: Shackleton’s Incredible Voyage* (1959); years later, in 2001, George Butler released a documentary about the expedition.

32

While researching a topic, a student has taken the following notes:

- The meter of a poem is the rhythmic structure or pattern of accents in its lines.
- Alliterative meter is structured by a pattern of repeated sounds.
- Quantitative meter is structured by a pattern of long and short syllables.
- The Old English poem *Widsith* uses an alliterative meter.
- The Sanskrit poem *Meghadūta* uses a quantitative meter.

The student wants to emphasize a difference between the meters of the two poems. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- The poem *Widsith* is written in Old English, but *Meghadūta* is written in Sanskrit.
- Alliterative meter is a pattern of repeated sounds, but quantitative meter is a rhythmic structure or pattern of accents.
- The Sanskrit poem *Meghadūta* uses a quantitative meter, while the meter of the Old English poem *Widsith* uses a pattern of long and short syllables.
- The lines of the poem *Meghadūta* use a pattern of long and short syllables, whereas *Widsith*'s lines use a pattern of repeated sounds.

33

While researching a topic, a student has taken the following notes:

- Leaders of the Province of Guatemala proclaimed independence for Central America from the Spanish Empire on September 15, 1821.
- The accompanying Declaration of Independence was written by Honduran scholar and politician José Cecilio del Valle.
- The 1812 Spanish Constitution had provided some degree of independence for Central America, but it was repealed by the Spanish king in 1814.
- Valle, a loyal advisor to the Spanish Empire's administrators in Central America, had long opposed independence.
- He changed his mind after Colonel Rafael del Riego's 1820 revolt, which demanded the return of rights lost in 1814.

The student wants to place the 1821 Declaration of Independence in the context of Valle's changing political beliefs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- Colonel Riego's revolt was the inspiration that Valle, a long-standing opponent of Central American independence, needed to change his political beliefs.
- Long an opponent of Central American independence, Valle changed his mind after an 1820 revolt and wrote the 1821 declaration.
- A change in Valle's political beliefs that occurred when the Spanish king repealed the 1812 constitution led to Valle writing Central America's Declaration of Independence.
- The writing of Central America's Declaration of Independence may not have happened were it not for Colonel Riego's 1820 revolt.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**

Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

Ancient Inca inhabiting the Andes Mountains used terraces, rows of flattened land with built-in irrigation systems, to grow crops at different altitudes. This method of farming proved to be highly _____, as evidenced by the great number and variety of crops grown at that time.

Which choice completes the text with the most logical and precise word or phrase?

- A) chaotic
- B) uniform
- C) effective
- D) burdensome

2

Since the Hubble Space Telescope was launched into space in 1990, astronauts have needed to complete regular missions to repair the telescope and keep it working smoothly. Researchers hope that robots will soon be able to make these repairs. Employing robots instead of humans to make repairs will be helpful, as _____ astronauts to maintain the telescope can be expensive.

Which choice completes the text with the most logical and precise word or phrase?

- A) straightening
- B) forgetting about
- C) relying on
- D) reducing

3

There are many famous examples of election pollsters making inaccurate predictions in presidential elections. But neuroscientist and election pollster Sam Wang has said that these prediction failures should not lead campaigns to _____ election polling entirely. Polling is about more than just predicting the winner; throughout campaigns, it helps strategists identify where their efforts are most likely to be effective.

Which choice completes the text with the most logical and precise word or phrase?

- A) distort
- B) enact
- C) neglect
- D) supplement

4

Steiger Butte Drum, a family ensemble from the Klamath Tribes of the Pacific Northwest, collaborated with composer Michael Gordon to create *Natural History*, a work featuring traditional drumming and vocals alongside an orchestra and chorus. Steiger Butte Drum's participation is _____ to the piece: members not only contributed to its composition but also must be included in all performances.

Which choice completes the text with the most logical and precise word or phrase?

- A) tangential
- B) subsequent
- C) analogous
- D) integral

5

The following text is from a 1955 translation of Samuel Beckett's 1951 novel *Molloy* (translated by the author and Patrick Bowles). In the text, Molloy has arrived at the town ramparts, an elevated walkway atop the city walls.

And having cleared the ramparts I had to confess the sky was clearing, prior to its winding in the other shroud, night. Yes, the great cloud was ravelling, discovering here and there a pale and dying sky, and the sun, already down, was manifest in the livid tongues of fire darting towards the zenith, falling and darting again, ever more pale and languid, and doomed no sooner lit to be extinguished.

©1955 by Grove Press, Inc.

As used in the text, what does the word "manifest" most nearly mean?

- A) Realized
- B) Perceptible
- C) Situated
- D) Dwindling

6

In the past, historians who wanted to examine Frederick Douglass's diary and other personal papers had to visit the Library of Congress in Washington, DC, to view them on microfilm (film containing scaled-down reproductions of documents). But traveling to the library often added time and costs to research projects. Now, by going to the library's website, researchers can access digitized versions of Douglass's papers without physically going anywhere.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It gives information about a famous person.
- B) It explains the meaning of a word.
- C) It describes a debate among historians.
- D) It summarizes an unexpected finding.

7

Following the eradication of the gray wolf in Yellowstone National Park in 1926, the population of elk—a primary prey of the gray wolf—exceeded a healthy size for the park’s ecosystem. Elk overpopulation led to overgrazing of areas that a multitude of other animals relied on for food and shelter. As scientists began to see how essential the gray wolf was to the Yellowstone food chain, ecological restoration strategies were employed to reintroduce the gray wolf to the park in 1996. The rebound effect in the park’s natural ecosystem was noticed almost immediately.

Which choice best describes the overall structure of the text?

- A) It summarizes a problem that developed in Yellowstone National Park in the 1920s and then offers potential solutions to that problem.
- B) It mentions the elimination of the gray wolf from Yellowstone National Park and then explains why the wolf was eventually restored to the park.
- C) It presents a claim about the health of the Yellowstone National Park gray wolf population and then gives specific examples to support that claim.
- D) It explains why Yellowstone National Park allowed the eradication of the gray wolf and then discusses the consequences of reintroducing the wolf to the park.

8

A team of researchers discovered that Matabele ants can identify an infected wound in a member of the colony and then treat the infection by covering the wound with antimicrobial secretions that the ants produce. The team found that the mortality rate for Matabele ants with infected injuries was reduced by 90% with this treatment, and they are hopeful that this discovery could aid in the development of new antibiotics for human use.

Which choice best describes the overall structure of the text?

- A) It summarizes research findings on Matabele ants and then identifies an area for further research.
- B) It introduces a study of Matabele ants and then explains the research methods used in the study.
- C) It describes unique properties of Matabele ants and then speculates on how those properties evolved.
- D) It identifies an issue concerning Matabele ants and then proposes a solution to address the issue.

9

Built in the 1970s, Raccoon Mountain is a pumped-storage hydropower facility (a “water-battery”) located in the United States along the Tennessee River. When energy demand is low, excess power from the regional electric utility’s nuclear plants is used to pump water (from a lower reservoir filled from the Tennessee River) up a shaft to the summit lake, where the water is stored as gravitational potential energy. When energy demand peaks, the water drains down from the summit lake, spinning turbines and generating upward of 1,700 megawatts of power—enough to power one million homes for twenty hours.

Which choice best states the main purpose of the text?

- A) To point out the differences between two methods of energy generation
- B) To explain the basics of how a specific energy technology works
- C) To encourage regional electric utilities to build energy storage facilities
- D) To discuss the benefits of a new energy technology

10

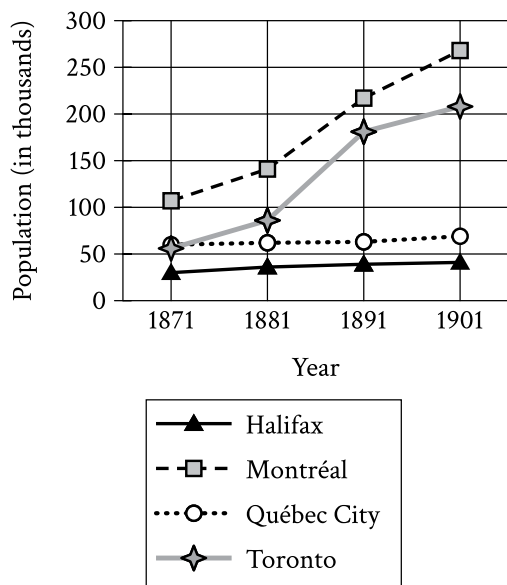
Ruth Asawa was an accomplished artist who worked in many art forms, including her unique tied-wire sculptures, but she was dedicated to more than the creation of art. Asawa also wanted to bring art to children in her hometown of San Francisco, California. To that end, in 1968 she cofounded the Alvarado School Arts Workshop, which brought works of art and artists into public schools, and in 1982 she helped found a San Francisco public arts high school, which was later named after her.

Which choice best states the main idea of the text?

- A) Asawa inspired many other artists to share their work with students in public schools.
- B) Asawa’s unique sculptures were appreciated more by local art communities than they were nationwide.
- C) Asawa’s interest in art education prompted her to create art programs for students in San Francisco.
- D) Asawa left a promising career as a sculptor to work as an art teacher in San Francisco schools.

11

Census Data for Four Canadian Cities,
1871–1901



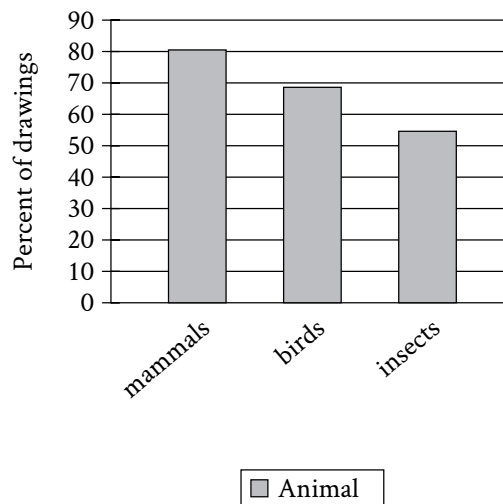
The first national census in Canada was conducted in 1871. The graph shows the population sizes of some of the largest Canadian cities from each of the first four national censuses, which occurred every ten years.

According to the graph, which city had the largest population in 1891?

- A) Toronto
- B) Montréal
- C) Québec City
- D) Halifax

12

Percent of Drawings
Containing an Example of
Each Animal Group

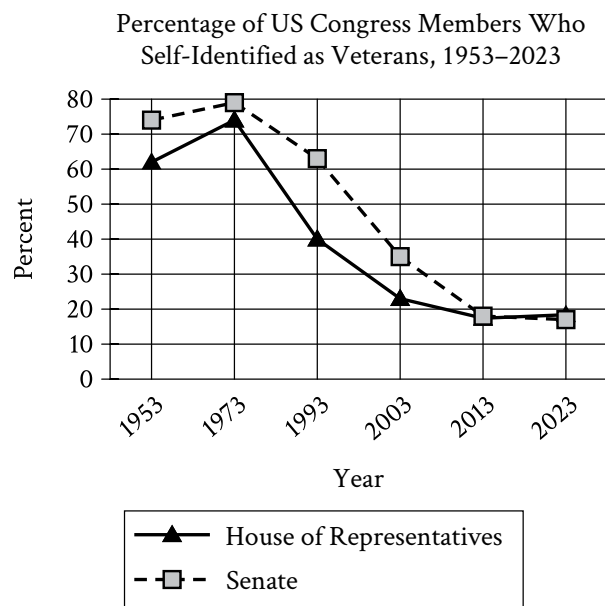


Researchers Kate Howlett and Edgar Turner asked 401 schoolchildren in the United Kingdom to draw and label local wildlife. The researchers suggest that the children may be more aware of mammals than of other animals in their environments.

Which choice best describes data from the graph that support the researchers' conclusion?

- A) There were about as many drawings with an example of an insect as there were with an example of a bird.
- B) The fewest number of drawings contained an example of a bird.
- C) More of the drawings had an example of an insect than an example of a mammal.
- D) More of the drawings had an example of a mammal than either an example of a bird or an example of an insect.

13



Since the nation’s earliest days, both houses of the United States Congress have counted many military veterans among their elected officials, including Hiram L. Fong (who served in the US Army Air Force) and Ralph Harold Metcalfe Sr. (who served in the US Army). A research institute gathered historical data about congressional members who have reported past military experience and determined that _____

Which choice most effectively uses data from the graph to complete the sentence?

- A) from 2003 to 2023, the House of Representatives had a higher percentage of those members than the Senate did.
- B) from 1953 to 2003, those members constituted a majority in both houses of Congress.
- C) the percentage of those members decreased substantially in both houses of Congress from 1973 to 2013.
- D) the percentage of those members remained fairly consistent, regardless of house, from 1993 to 2023.

14

“Aunt Sue’s Stories” is a 1926 poem by Langston Hughes. In the poem, the speaker indicates that the stories Aunt Sue tells are based on Aunt Sue’s personal experiences, saying that _____

Which choice most effectively uses a quotation from “Aunt Sue’s Stories” to illustrate the claim?

- A) dark shadows “cross and recross / Aunt Sue’s stories.”
- B) a listening child “knows that Aunt Sue / Never got her stories out of any book at all, / But that they came / Right out of her own life.”
- C) the stories are told during “Summer nights on the front porch.”
- D) the people in the stories “Mingle themselves softly / In the flow of old Aunt Sue’s voice, / Mingle themselves softly.”

15

Olms are salamanders that live in underwater caves. Scientists once thought that olms stay in their caves all their lives. However, Raoul Manenti and team claim that olms regularly come to the surface to perform important activities such as finding food.

Which finding, if true, would most strongly support the underlined claim?

- A) Researchers determine that olms don’t breed often.
- B) Researchers confirm that olms live in only a few cave systems.
- C) Researchers discover that earthworms from surface soils are a major part of olms’ diet.
- D) Researchers learn that olms’ brains differ from other salamanders’ brains.

16

In Diné (Navajo) culture, *ikaah* (sandpaintings) are created and then erased as part of sacred healing ceremonies lasting no more than a few days, but Diné *hataalii* (chanter, healer, cultural guide) Fred Stevens developed fixatives to preserve desacralized sandpaintings. While on a US-sponsored cultural ambassadorial trip in Europe and the Americas in the 1960s, Stevens produced several such sandpaintings and gave them to cultural institutions. This may seem in tension with the role of cultural ambassador—how could static objects authentically represent an inherently ephemeral and dynamic practice?—but such a view is itself overly object-focused and neglects how Stevens strove to convey exactly those characteristics of *ikaah* as a cultural practice.

Which quotation from an art historian would most directly support the claim made in the text?

- A) “While Stevens’s ambassadorial sandpaintings are undoubtedly educational for audiences that have little exposure to Diné culture, they should not be confused with authentic *ikaah*, which cannot be extricated from a practice that is intentionally and necessarily transitory nor condensed into a single persistent object.”
- B) “The most compelling way to reconcile the apparent tension between the temporary and performative nature of *ikaah* and the persistent and static nature of Stevens’s ambassadorial sandpaintings is to recognize that Stevens was an ambassador not only of Diné culture but of a US art culture that tended to value permanent works over ephemeral ones.”
- C) “Stevens’s ambassadorial sandpaintings are best understood not as self-contained objects but as reminders of the public creations of the sandpaintings, during which Stevens conducted appropriate *ikaah* rituals and encouraged viewers to closely track his movements and subtle shifts in the sand throughout the process.”
- D) “It is important to recall Stevens’s own actions as a gift-giver—a temporary and performative role—of the sandpaintings, for by taking those actions, Stevens publicly enacted the transmission of knowledge of both traditional *ikaah* practices and cutting-edge chemistry (in the form of the preservative additives), a hybrid that reflects the dynamism of Diné culture.”

17

In Switzerland, the white fuzzy mountain flowers known as edelweiss are widely treated as a symbol of strength and courage. Although edelweiss can thrive in extreme conditions, they aren't notably tougher or harder to reach than other mountain flowers growing in the Swiss Alps. Historian Tobias Scheidegger has shown that the popular view of the flowers originated in the mid-1800s when mountain climbing became popular in Switzerland. Mountain climbers spread the idea that the flowers grew only in steep, icy terrains that were dangerous to climb to. Scheidegger says that these claims were self-interested. He suggests that mountain climbers presented edelweiss in this way in order to _____

Which choice most logically completes the text?

- A) make themselves appear brave and strong for being able to climb to difficult places where edelweiss supposedly grew.
- B) encourage more flower enthusiasts to explore the Swiss Alps.
- C) share their observations about the unusual characteristics of edelweiss with scientists.
- D) prove that edelweiss were more common in the Swiss Alps than in other mountain regions in Europe.

18

The ratio of methane to other atmospheric constituents—represented by a measure called the methane mole fraction—influences a variety of meteorological phenomena, notably precipitation and humidity. For Titan, Saturn's largest moon, the observational data that exist are too sparse and discrepant to fully constrain the range of the methane mole fraction at various atmospheric levels. Juan Lora and colleagues point out that outputs of the IPSL atmospheric model of Titan, which track closely to observations in some respects, reflect how the model's developers responded to this challenge: by prescribing a uniform methane mole fraction for the lowest level of the atmosphere. It is therefore important to note that _____

Which choice most logically completes the text?

- A) some disagreements between the model's simulations of Titan's precipitation and humidity and the moon's actual precipitation and humidity are to be expected.
- B) further observations of Titan may clarify the moon's methane mole fraction sufficiently for the model to employ a single value rather than a range.
- C) even though the model's outputs sometimes agree with observational data, Titan's real methane mole fraction is likely higher than the methane mole fraction used in the model.
- D) inconsistencies across the model's simulations of Titan's precipitation and humidity could be attributable to variations in the moon's methane mole fraction.

19

Ganga is a folk singing tradition that originated in the Dinaric Alps in southern Europe. Ganga singers sing different melodies at the same time. The clashing notes can echo a long way across the mountains, which is why _____

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) ganga has been used as a communication method?
- B) ganga has been used as a communication method.
- C) has ganga been used as a communication method.
- D) has ganga been used as a communication method?

20

A popular suite of mapping and spatial analysis software, ArcGIS enables cartographers like Karachi Cartography founder Namra Khalid _____ maps by analyzing and arranging raw geospatial data.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) create
- B) to create
- C) creating
- D) created

21

The part of a compound that determines the compound's color is _____ the chromophore. One example of a chromophore is hemoglobin, which gives human blood its red color.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) called,
- B) called
- C) called—
- D) called;

22

In rural Minnesota, ecologist Leroy Walston conducted a study to determine whether seeding solar panel fields with wildflowers could bolster pollinator populations in nearby food crops. Walston's findings indicate that—assuming solar panel installers' _____ practice has the potential to increase the number of native bees in crops near solar fields throughout the Midwest by up to 20 percent.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) cooperation, this
- B) cooperation—this
- C) cooperation: this
- D) cooperation this

23

Key-value pairs are an important aspect of JavaScript Object Notation (JSON), an electronic file format for storing and transmitting data. Keys function as labels, while values contain the actual information. In a JSON file storing data about fire belly _____ instance, you could encounter a key such as “species” with the associated value of “*Cynops orientalis*.”

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) newts. For
- B) newts, for
- C) newts: for
- D) newts; for

24

Most of the ice found on Earth is ice Ih, distinguished by a crystalline structure in which molecules form a hexagonal pattern. Amorphous ice, on the other hand, constitutes most of the ice in the ultrafrigid environment of outer space. Defined by a disorganized molecular structure, _____

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) ice Ih contains crystals, whereas amorphous ice, which lacks the thermal energy to form them, does not.
- B) amorphous ice lacks the thermal energy to form the crystals found in ice Ih.
- C) the lack of thermal energy in amorphous ice explains its inability to form the crystals found in ice Ih.
- D) ice Ih differs from amorphous ice in that it possesses the thermal energy to form crystals.

25

Recordings of electrical activity in the brain, _____ increased activity in brain areas associated with suppressing motor functions.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) electrograms show that while responding to hypothetical match scenarios, the most highly skilled soccer players have
- B) the most highly skilled soccer players responding to hypothetical match scenarios have electrograms that show
- C) responses to hypothetical match scenarios show that the most highly skilled soccer players have electrograms with
- D) hypothetical match scenario responses show that the most highly skilled soccer players captured in electrograms have

26

If an animal can recognize itself in a reflective surface (“the mirror test”), it is considered to have self-awareness. In a recent study, scientists _____ for evidence of self-awareness in snakes, species that rely primarily on olfactory rather than visual processing, adapted the test to foreground smell, modifying the scent trails of North American eastern garter snakes and African ball pythons.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) searched
- B) searching
- C) were searching
- D) have searched

27

The traditional process of Turkish paper marbling (ebru) generally proceeds like this: First, the artisan fills a shallow tray with a water bath solution. Next, the artisan adds inks or paints to the solution, which can then be manipulated into intricate designs. _____ the artisan slips paper in and out of the liquid, transferring the design onto the paper.

Which choice completes the text with the most logical transition?

- A) Actually,
- B) Therefore,
- C) Nevertheless,
- D) Finally,

28

While studying jorō spiders, a large species originally from East Asia, University of Georgia researchers wondered if the spiders’ rapid spread throughout the southeastern US was a result of aggressive behavior. _____ they discovered that jorō spiders are gentle giants who react to even minor disturbances by “freezing” in place for an hour or more.

Which choice completes the text with the most logical transition?

- A) Therefore,
- B) Instead,
- C) For example,
- D) In other words,

29

Upper-atmospheric jets—phenomena whereby electrical discharges cause narrow cones of blue light to briefly burst upward from the tops of thunderclouds—have been observed reaching the ionosphere. The extreme altitudes involved (the ionosphere begins about 80 km above Earth) mark these gigantic jets as outliers; _____ the majority of jets reach heights of only 20 to 50 km.

Which choice completes the text with the most logical transition?

- A) nevertheless,
- B) consequently,
- C) indeed,
- D) in addition,

30

In Annie Dillard’s *Pilgrim at Tinker Creek*—where, early on, the author marvels at a single goldfish’s delicate fins but later winces when imagining a horde of goldfish laying and eating their own eggs—Dillard struggles to reconcile the complicated juxtapositions of the natural world. _____ nature’s mesmerizing intricacy and pitiless harshness prove inextricably linked for Dillard, like “two branches of the same creek.”

Which choice completes the text with the most logical transition?

- A) To that end,
- B) Ultimately,
- C) Moreover,
- D) Hence,

31

While researching a topic, a student has taken the following notes:

- In a 2020 study, researchers in California investigated how many potential nesting sites female wood ducks visited during the nesting season.
- The researchers placed nest boxes throughout the survey area and tagged 138 female wood ducks with radio frequency ID trackers.
- These trackers recorded how many nest boxes each duck visited.
- 67 ducks (48.5%) visited only one nest box.
- 18 ducks (13.0%) visited 10 or more nest boxes.
- Younger ducks were more likely to visit multiple nest boxes.

The student wants to present the methods used in the 2020 study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) By recording how many nest boxes each duck visited, the researchers discovered that only a relatively small percentage (13.0%) of the ducks visited 10 or more nest boxes.
- B) After tracking how many nest boxes the 138 wood ducks visited, the researchers found that the younger ducks tended to visit more nest boxes than the older ducks.
- C) The researchers tagged 138 female wood ducks with radio frequency ID trackers and recorded how many nest boxes each duck visited during the nesting season.
- D) The researchers investigated each nesting site for signs that it had been visited by the female wood ducks.

32

While researching a topic, a student has taken the following notes:

- Pedestrian malls are outdoor streets in a city or town where vehicle traffic is prohibited.
- Many pedestrian malls were built in the 19th and 20th centuries in Europe and Asia.
- Qianmen Dajie is a famous pedestrian mall in Beijing.
- It has existed since the Ming dynasty (1368–1644 CE).
- Rue Mouffetard is a famous pedestrian mall in Paris.
- It has existed since the mid-Roman Empire (117–235 CE).

The student wants to emphasize a similarity between the ages of the malls. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Qianmen Dajie pedestrian mall has roots as far back as the Ming dynasty; likewise, Rue Mouffetard has existed for centuries.
- B) Both Qianmen Dajie and Rue Mouffetard are famous pedestrian malls, the former in Beijing and the latter in Paris.
- C) Qianmen Dajie and Rue Mouffetard are pedestrian malls, outdoor streets closed to vehicle traffic.
- D) Qianmen Dajie and Rue Mouffetard are examples of pedestrian malls, which proliferated in Europe in the 19th and 20th centuries.

33

While researching a topic, a student has taken the following notes:

- Modularity of mind is the notion that the mind is at least partly composed of innate neural structures (modules) that perform fast, necessary tasks.
- 1983: cognitive scientist Jerry A. Fodor hypothesized that low-level cognitive systems (e.g., perception, language) are modular.
- In Fodorian modularity, high-level systems (e.g., reasoning) are not modular.
- 2003: cognitive scientist Peter Carruthers proposed the massive modularity hypothesis (MMH).
- MMH expands modularity to include all cognitive systems.

The student wants to compare Fodor's hypothesis with Carruthers's. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) In considering some but not all cognitive systems modular, Fodorian modularity is not as expansive in its definition of modularity as MMH is.
- B) Following Fodor's 1983 hypothesis, Carruthers proposed that modularity of mind includes all cognitive systems.
- C) The hypotheses of Fodor and Carruthers differ in whether they consider low-level cognitive systems, such as perception and language, modular.
- D) In 2003, Carruthers proposed the massive modularity hypothesis, disagreeing with Fodor's earlier hypothesis that the mind is composed of innate neural structures.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**

Math

27 QUESTIONS

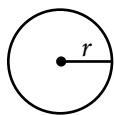
DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

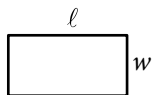
Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

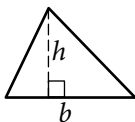
REFERENCE

$$A = \pi r^2$$

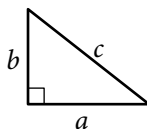
$$C = 2\pi r$$



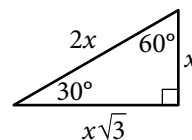
$$A = \ell w$$



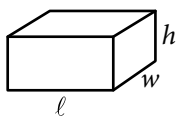
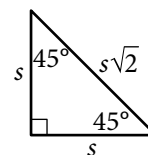
$$A = \frac{1}{2}bh$$



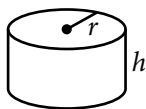
$$c^2 = a^2 + b^2$$



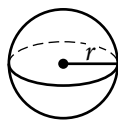
Special Right Triangles



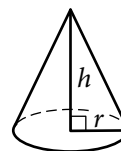
$$V = \ell wh$$



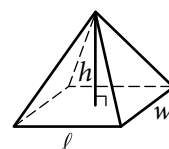
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

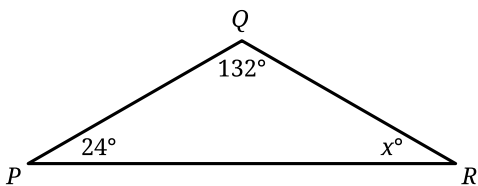
The sum of the measures in degrees of the angles of a triangle is 180.

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction ($\frac{7}{2}$) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1



Note: Figure not drawn to scale.

In the triangle shown, $PQ = QR$. What is the value of x ?

- A) 156
- B) 66
- C) 48
- D) 24

2

$$4x + 1 = 33$$

Which equation has the same solution as the given equation?

- A) $4x = 32$
- B) $4x = 5$
- C) $4x = 1$
- D) $4x = -32$

3

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7 and passes through the point $(0, 5)$. Which equation defines f ?

- A) $f(x) = 5x$
- B) $f(x) = 35x$
- C) $f(x) = 7x + 5$
- D) $f(x) = 12x + 5$

4

$$8x^2 - 40 = 32$$

What is the positive solution to the given equation?

- A) 3
- B) 4
- C) 9
- D) 72

5

A total of 50 children attended a summer camp and were offered 4 types of sandwiches. The table shows the number of children who chose each type of sandwich.

Type of sandwich	Number of children
Turkey	15
Chicken	23
Ham	3
Vegetarian	9
Total	50

If one of these children is selected at random, what is the probability of selecting a child who chose a vegetarian sandwich?

- A) $\frac{9}{100}$
 B) $\frac{9}{50}$
 C) $\frac{1}{4}$
 D) $\frac{9}{10}$

6

Amara grows cherry tomatoes in her backyard. This year, she harvested 750 cherry tomatoes and gave 10% of them to her neighbor. How many of the harvested cherry tomatoes did Amara give to her neighbor?

7

$$x + y = 125$$

$$x + y + y = 155$$

The solution to the given system of equations is (x, y) . What is the value of y ?

8

In a chess tournament, each participant earns 1 point for each game the participant plays that ends in a draw and 3 points for each game the participant wins. A certain participant in this tournament has earned 41 points. Which equation represents this situation, where d represents the number of games this participant has played that ended in a draw and w represents the number of games this participant has won?

- A) $d + 3w = 41$
 B) $3d + w = 41$
 C) $d + \frac{w}{3} = 41$
 D) $\frac{d}{3} + w = 41$

9

The function g is defined by $g(x) = \sqrt{x} + 300$.

What is the value of $g(x)$ when $x = 81$?

- A) 9
- B) 300
- C) 309
- D) 381

10

A cable provider wanted to know how many of its 30,000 customers would be interested in a new service plan. The provider selected 300 customers at random and asked each customer whether the customer would be interested in the new plan. Of those surveyed, 8 said they would be interested. Which of the following is the best estimate of the total number of customers who would be interested in the new service plan?

- A) 8
- B) 80
- C) 800
- D) 8,000

11

Which expression is equivalent to $64t^2s^3 - 56t^3s$?

- A) $4ts(16s^2 - 14ts)$
- B) $4ts(16t^2s^2 - 14t)$
- C) $4t^2s(16ts - 14s)$
- D) $4t^2s(16s^2 - 14t)$

12

$$\begin{aligned}x + 5 &= 14 \\ y &= 4x^2 + 4\end{aligned}$$

At what point (x, y) do the graphs of the equations in the given system intersect?

- A) (9, 324)
- B) (9, 328)
- C) (14, 4)
- D) (14, 788)

13

$$8x + 11y = 170$$

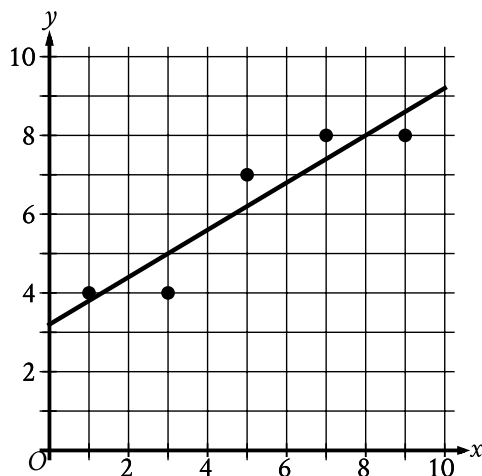
The equation gives the possible combinations of the number of 2009 premium grade Log Cabin Pennies, x , and the number of 1996 select grade Lincoln Pennies, y , in a collection that is worth a total of \$170. If there are 6 1996 select grade Lincoln Pennies in the collection, how many 2009 premium grade Log Cabin Pennies are in the collection?

14

The population of the town of Smithville doubled every 75 years from 1659 to 1959. The population of this town was 240,000 in 1959. What was the population of this town in 1659?

15

The scatterplot shows the relationship between x and y . A line of best fit is also shown.



Which of the following is closest to the slope of this line of best fit?

- A) 0.60
- B) 2.50
- C) 7.80
- D) 8.00

16

Which quadratic equation has exactly one distinct real solution?

- A) $(x + 15)^2 = 0$
- B) $(x + 15)^2 = -45$
- C) $(x + 15)^2 = 45$
- D) $(x + 15)^2 = 135$

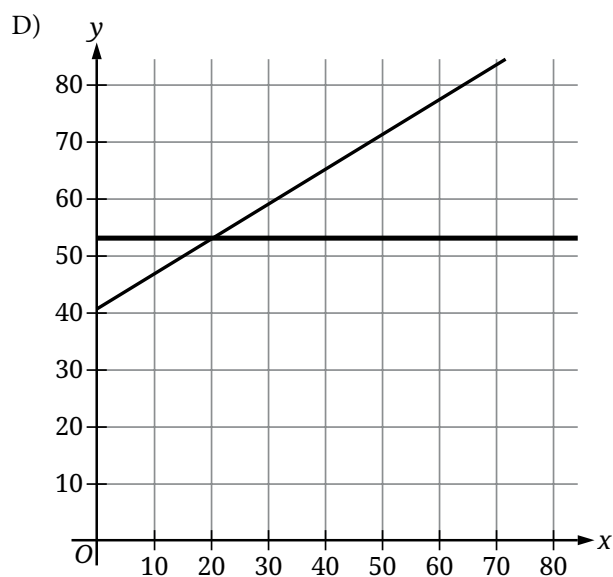
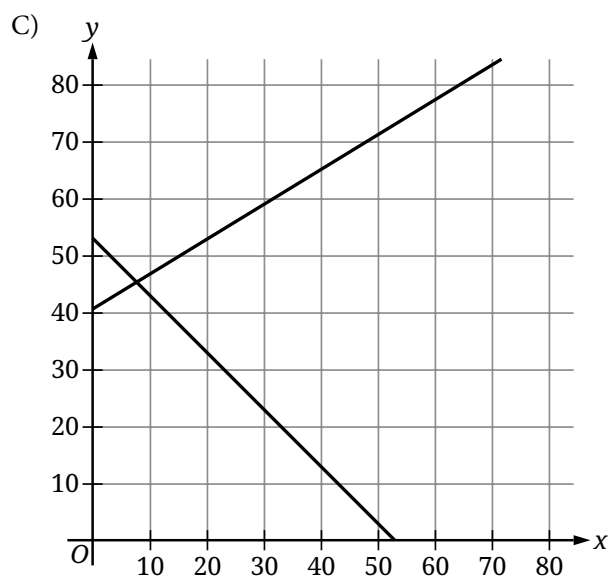
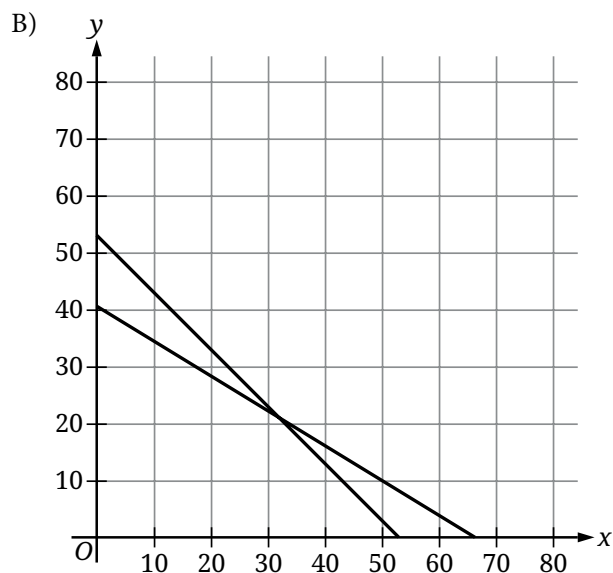
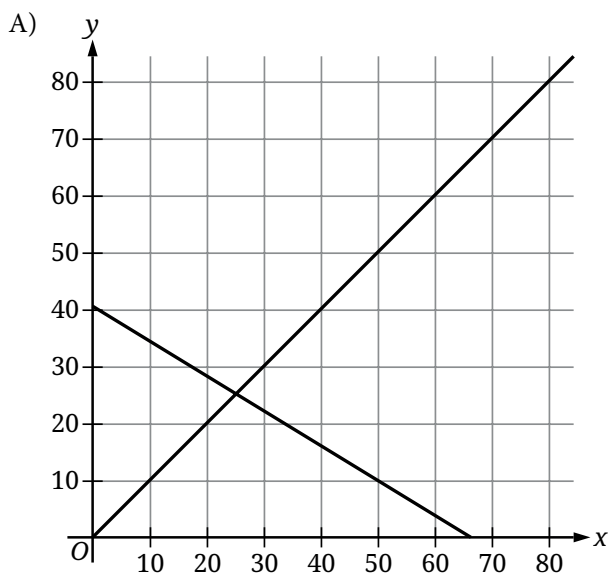
17

The cost to rent a bus from Company X is \$950 for the first 3 hours and an additional \$50 per hour for each hour after the first 3 hours. If the total cost to rent the bus from Company X for t hours, where $t > 3$, is \$1,150, which equation represents this situation?

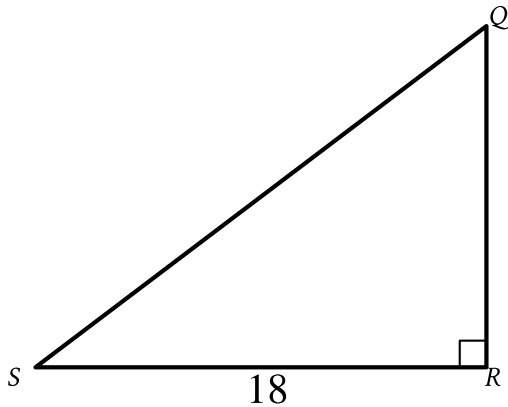
- A) $950(t - 3) + 50t = 1,150$
- B) $950(3t) + 50t = 1,150$
- C) $950 + 50(t - 3) = 1,150$
- D) $950 + 50(3t) = 1,150$

$$x + y = 53$$
$$11x + 18y = 730$$

The given equations represent the possible numbers of beach chairs, x , and umbrellas, y , rented at a park last month and the total spent, in dollars, to rent those beach chairs and umbrellas. Which of the following graphs represents this situation?



19



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

- A) $18 \cos Q$
- B) $18 \sin Q$
- C) $\frac{18}{\cos Q}$
- D) $\frac{18}{\sin Q}$

20

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 25$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

21

$$x^2 + 7x + 5 = 0$$

One solution to the given equation can be written as

$$x = \frac{-7 + \sqrt{k}}{2}, \text{ where } k \text{ is a constant. What is the}$$

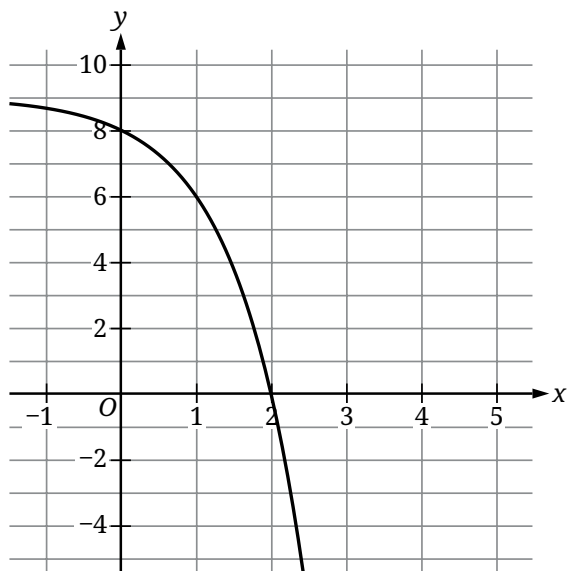
value of k ?

22

A scientist measured the lengths of 240 gray seals from Muskeget Island and 120 gray seals from Sable Island. The scientist determined that the mean length of the 240 gray seals from Muskeget Island was 88 inches and the mean length of the 120 gray seals from Sable Island was 94 inches. What was the mean length of all 360 gray seals the scientist measured for this study?

- A) 89
- B) 90
- C) 91
- D) 92

23



The graph of $y = f(x) + 4$ is shown. Which equation defines function f ?

- A) $f(x) = -3^x + 1$
- B) $f(x) = -3^x + 5$
- C) $f(x) = -3^x + 8$
- D) $f(x) = -3^x + 9$

24

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 140)^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

- A) $(-18x + 280)^\circ$
- B) $(-18x + 640)^\circ$
- C) $(18x - 280)^\circ$
- D) 180°

25

$$2x + 9y = 7$$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

- I. $3x + 13.5y = 10.5$
- II. $3x - 13.5y = 10.5$

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

26

A right rectangular prism has a base area of $24t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the rectangular prism is 15 cm. Which expression represents the surface area, in cm^2 , of the right rectangular prism?

- A) $48t + 160$
- B) $318t + 80$
- C) $1,968t + 80$
- D) $360t$

27

For a particular car, the linear function f gives the predicted power, in brake horsepower (bhp), for engine speeds between 1,000 revolutions per minute (rpm) and 6,000 rpm. According to this function, the car's predicted power is 433 bhp at an engine speed of 3,331 rpm and 600 bhp at an engine speed of 4,500 rpm. The equation $f(x) = \frac{1}{7}(x - a) + 433$ defines f , where x is the engine speed, in rpm, and a is a constant. What is the value of a ?

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**

Math

27 QUESTIONS

DIRECTIONS

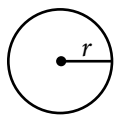
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

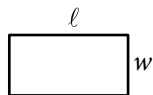
- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

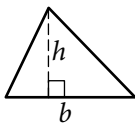


$$A = \pi r^2$$

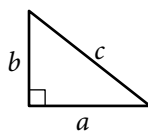
$$C = 2\pi r$$



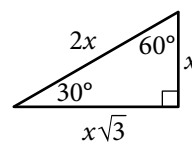
$$A = \ell w$$



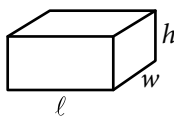
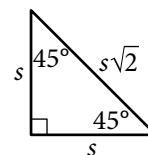
$$A = \frac{1}{2}bh$$



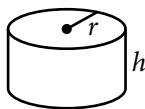
$$c^2 = a^2 + b^2$$



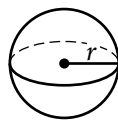
Special Right Triangles



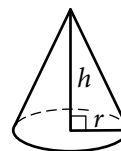
$$V = \ell wh$$



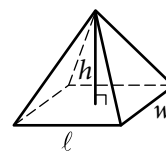
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction ($\frac{7}{2}$) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

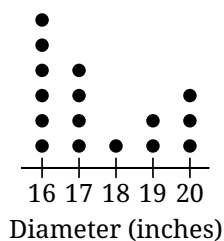
1

Which expression is equivalent to $6x + 5x + 4y$?

- A) $15x$
- B) $15y$
- C) $11x + 4y$
- D) $30x + 4y$

2

To study the characteristics of sea stars in a group of tide pools, researchers measured the diameter of the sea stars within the tide pools. The dot plot gives the diameter, to the nearest inch, of each of the sea stars in these tide pools.



Based on the dot plot, how many sea stars had a diameter, to the nearest inch, of 16 inches?

- A) 16
- B) 6
- C) 4
- D) 1

3

A rectangle has a length of 56 inches and a width of 28 inches. What is the area, in square inches, of the rectangle?

- A) 28
- B) 84
- C) 168
- D) 1,568

4

$$10x = 110$$

$$6x - 63 = y$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A) 63
- B) 11
- C) 10
- D) 3

5

The function f is defined by $f(x) = 9(2x + 3)$. For what value of x does $f(x) = 63$?

- A) 2
- B) 5
- C) 7
- D) 30

6

$$10x = 86$$

What value of x is the solution to the given equation?

7

$$y = 3,600(a)^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y , in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

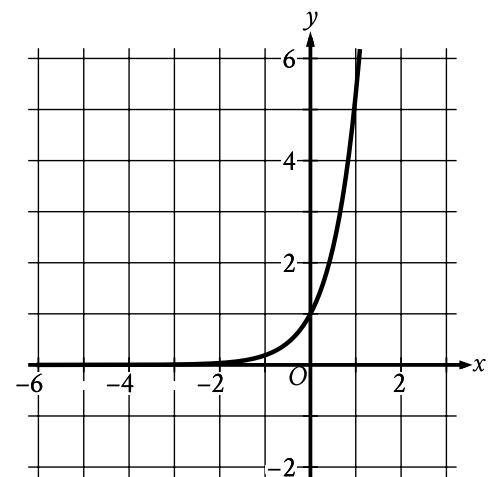
8

Leo goes to a packing store to buy containers and tape. Leo has \$15. Each container costs \$1.87 and each roll of tape costs \$2.40. Which inequality represents the relationship between the number of containers, c , and the number of rolls of tape, t , Leo can buy?

- A) $1.87c + 2.40t \leq 15$
- B) $1.87c + 2.40t \geq 15$
- C) $2.40c + 1.87t \leq 15$
- D) $2.40c + 1.87t \geq 15$

9

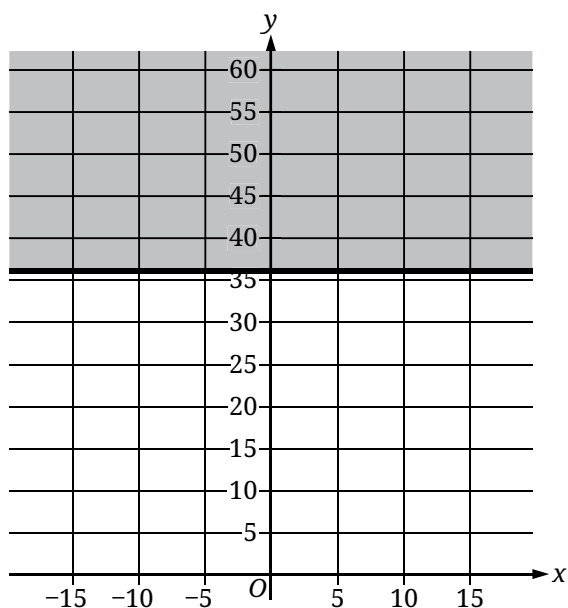
The graph of the function f is shown, where $y = f(x)$.



Which of the following best describes the function f ?

- A) Decreasing exponential
- B) Increasing exponential
- C) Decreasing linear
- D) Increasing linear

10



The shaded region shown in the graph represents all the solutions to which inequality?

- A) $x \leq 36$
- B) $x \geq 36$
- C) $y \leq 36$
- D) $y \geq 36$

11

There are 240 players in a tennis competition that includes 4 rounds of matches. Each player in the competition will play a match against another player in round 1. At the end of each round, the player who loses the match is eliminated and the player who won the match advances to the next round to play a match against another winning player. Which equation gives the number of players, p , eliminated at the end of round r , where $r \leq 4$?

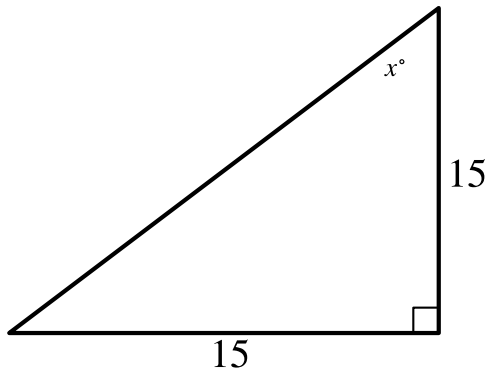
- A) $p = 15\left(\frac{1}{2}\right)^r$
- B) $p = 15(2)^r$
- C) $p = 240\left(\frac{1}{2}\right)^r$
- D) $p = 240(2)^r$

12

Line k is defined by $y = 6x + 4$. Line j is parallel to line k in the xy -plane and passes through the point $(0, 5)$. Which equation defines line j ?

- A) $y = 6x + 5$
- B) $y = -5x + 5$
- C) $y = -6x + 5$
- D) $y = 5x + 5$

13



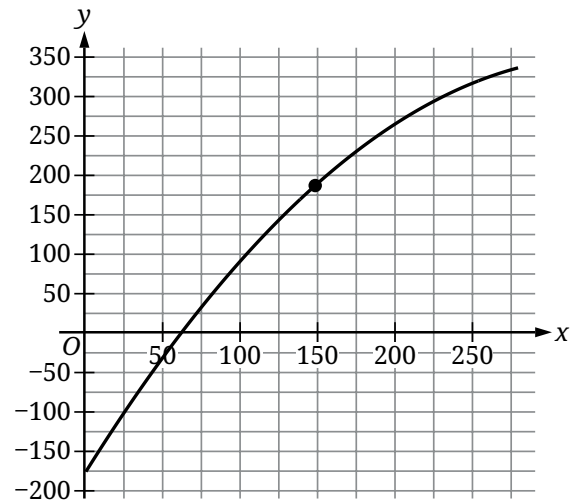
Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

14

What is the radius of the circle in the xy -plane defined by $(x + 2)^2 + (y + 5)^2 = 169$?

15



The graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(149.02, 186.05)$?

- A) A normal paraffin with a molecular weight of 186.05 grams per mole has an estimated boiling point of 149.02 degrees Celsius.
- B) A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.
- C) The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.
- D) The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

16

For the polynomial function f , the graph of $y = f(x)$ in the xy -plane passes through the points $(-5, 0)$, $(1, 0)$, and $(4, 0)$. Which of the following must be a factor of $f(x)$?

- A) $x + 1$
- B) $x + 4$
- C) $x - 1$
- D) $x - 5$

17

x	$g(x)$
1	32
2	28
3	24
4	20

For the linear function g , the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b ?

- A) 4
- B) 16
- C) 32
- D) 36

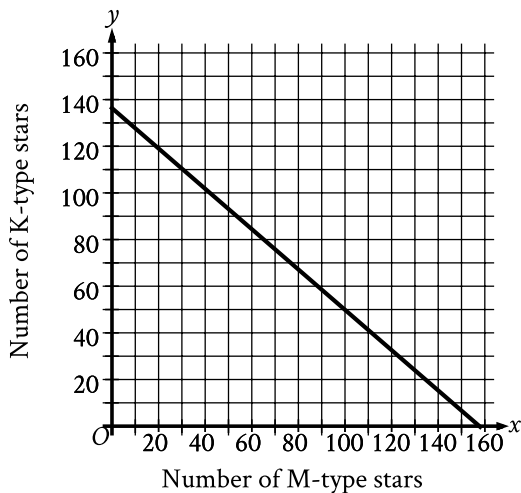
18

x	24	30	32
$f(x)$	-8	-8	8

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

- A) $f(x) = (x - 24)(x - 30) + 4$
- B) $f(x) = (x - 24)(x - 30) - 8$
- C) $f(x) = (x - 8)(x - 32) + 32$
- D) $f(x) = (x - 8)(x - 32) - 32$

19



A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is 127,882 quettagrams. The graph shown models the possible combinations of the number of M-type stars, x , and K-type stars, y , that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

- A) 811
- B) 938
- C) 51,904
- D) 75,978

20

$$\sqrt[3]{p^2} = t^{\frac{9}{7}}$$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{3n-1}$, where n is a constant, what is the value of n ?

21

The number a is 55% less than the number b . The number b is 320% greater than 160. What is the value of a ?

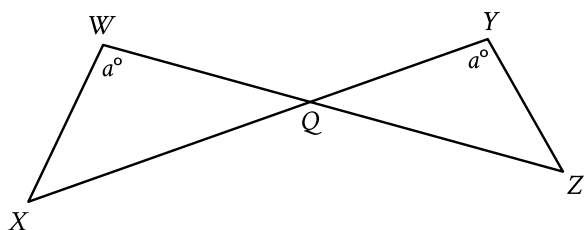
22

$$f(x) = -6x^2 + 60x - 126$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the maximum value of the function as a constant or coefficient?

- A) $f(x) = -6x^2 + 42x + 18x - 126$
- B) $f(x) = -6x(x - 7) + 18(x - 7)$
- C) $f(x) = -6(x - 5)^2 + 24$
- D) $f(x) = -6(x - 7)(x - 3)$

23



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

- A) 18
- B) 36
- C) 120
- D) 200

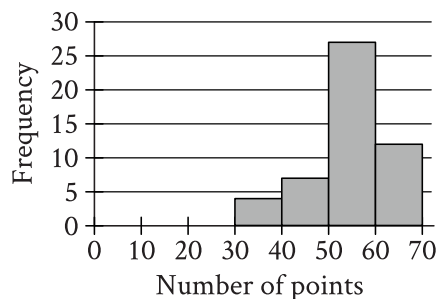
24

$$52(x^3 + 64)(x^4 - 81) = 0$$

How many distinct real solutions does the given equation have?

- A) Exactly two
- B) Exactly three
- C) Exactly five
- D) Exactly seven

25



The histogram summarizes data set A, which represents the number of points per player earned by 50 players of a game. A new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. Which of the following must be true?

- I. The median number of points per player for data set B is less than the median number of points per player for data set A.
- II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

26

In triangle XYZ , the measure of angle X is 90° . Point W lies on segment YZ , and segment WX is perpendicular to segment YZ . The length of segment WY is 572, and the length of segment WX is 429. What is the value of $\tan Z$?

- A) $\frac{3}{5}$
- B) $\frac{3}{4}$
- C) $\frac{4}{5}$
- D) $\frac{4}{3}$

27

An area of 46.00 square nautical miles is equivalent to k square kilometers. To the nearest tenth, what is the value of k ? (1 nautical mile = 1.852 kilometers)

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**

The SAT[®]

Practice Test #11

ANSWER EXPLANATIONS

These answer explanations are for students taking the digital SAT in nondigital format.



Reading and Writing

Module 1

(33 questions)

QUESTION 1

Choice A is the best answer because it most logically completes the text's discussion of Ezra Pound's poetry. In this context, "comprehend" would mean to understand or grasp the meaning of something. The text describes Pound's poetry as "dense" and "experimental," with many references that readers may not understand. The text also states that many readers may not even be able to identify a poem's subject. This context clearly suggests that readers struggle to understand Pound's poetry as a result of its complexity and obscure references.

Choice B is incorrect. To say that Pound's poetry is "hard to dislike" would suggest that people generally like it, but the text doesn't focus on how likeable or enjoyable the poems are; rather, it focuses on how challenging many readers find it to understand them. **Choice C** is incorrect. To say that Pound's poetry is "hard to interrupt" would mean that it is hard to stop or break into it while it's being read or recited, but the text doesn't discuss any such breaks; instead, the text focuses on the challenge of understanding the content of Pound's poems. **Choice D** is incorrect. In this context, "overlook" would mean look past or fail to notice, so to say that Pound's poetry is "hard to overlook" would mean that it is very likely to be noticed. The text focuses on the difficulty people have in understanding Pound's poems when reading them, not how noticeable his poetry is in the first place.

QUESTION 2

Choice C is the best answer because it most logically completes the text's discussion of Bali's *subak* water management system. In this context, "interconnected" would mean joined or linked together. The text describes the various components of the subak system—"terraces, canals, and water temples"—and then states that they are "joined together into a single cohesive unit." This context indicates that these elements are connected with one another in an integrated system.

Choice A is incorrect because “outmoded” would mean outdated or no longer in use. While the text does indicate that the subak system has “rich cultural, philosophical, and historical significance,” the text doesn’t suggest that the subak system is obsolete or not currently used. *Choice B* is incorrect because “informal” would suggest that the subak system lacks structure or organization. The text contradicts this by describing a complex system with specific components (“terraces, canals, and water temples”) that are “joined together into a single cohesive unit,” indicating formal organization rather than informality. *Choice D* is incorrect because “optional” would suggest that the various elements of the subak system are not necessary or required. The text indicates the opposite by stating that these elements are “joined together into a single cohesive unit,” suggesting that each component plays a necessary role in the functioning of the integrated system.

QUESTION 3

Choice D is the best answer because it most logically completes the text’s discussion of Georgi Vladimov’s novel *Faithful Ruslan*. In this context, “suppress” means to prevent something, such as a book, from being accessed by the public. The text indicates that the novel was circulated secretly within the Soviet Union, a stratagem that was necessary because of the government’s efforts to suppress the book, or prevent it from being circulated and read.

Choice A is incorrect because in this context, “replicate” would mean to copy or reproduce a book, and this meaning wouldn’t make sense in the text’s discussion of *Faithful Ruslan*. If the Soviet government had attempted to replicate Vladimov’s novel, there would be no reason for copies to circulate “in secret,” since the government would be actively making more copies available, not restricting access. *Choice B* is incorrect. Although a repressive government might well “critique” a novel, or evaluate it negatively, simply critiquing one wouldn’t necessarily lead to copies of it circulating “in secret.” Instead, the text suggests that the Soviet government took decisive action against *Faithful Ruslan* that necessitated its underground circulation and did not merely offer criticism of the novel. *Choice C* is incorrect because “import” means to bring a product, such as a book, into a country from elsewhere. However, the text suggests that copies of *Faithful Ruslan* were already present in the Soviet Union and that the government was trying to restrict their circulation rather than increasing access to the novel by importing copies from another country.

QUESTION 4

Choice D is the best answer because it most logically completes the text’s discussion of the initial spread of silk beyond China. In this context, “contradicted” means opposed or challenged. The text indicates that new archaeological evidence about silk use in South Asia has done something to a long-held thought about the initial spread of silk. According to the text, the new evidence reveals that people used silk in South Asia 1,000 years before the second century CE, which directly challenges the previous belief that silk first spread beyond China in the second century CE.

Choice A is incorrect because the text indicates that an archaeological finding about silk use has done something to a long-held thought about the initial spread of silk, and it wouldn't make sense to suggest that the finding could have "investigated," or examined, a belief; evidence can't examine anything. *Choice B* is incorrect because the text indicates that an archaeological finding about silk use has done something to a long-held thought about the initial spread of silk, and it wouldn't make sense to suggest that the finding could have "misinterpreted," or incorrectly understood, a belief; evidence can't try to understand anything. *Choice C* is incorrect. In this context, "anticipated" would mean expected or predicted. The text indicates that a new archaeological finding about silk use has done something to a long-held thought about the initial spread of silk, and a new finding can't predict a belief that is already established.

QUESTION 5

Choice A is the best answer because it most logically completes the text's discussion of Scherezade García's mural *Blame It on the Bean: The Power of Coffee*. In this context, "unassuming" means ordinary or unremarkable. The text draws a contrast between a "casual description" of the mural's ordinary location (a coffee shop) and subject matter (three women drinking coffee)—details that would make it seem relatively unambitious—and a characterization of the painting as a "dynamic meditation" on gender and colonialism deserving of "serious attention." This context suggests that the mural has significant depth and complexity, but a superficial description of it might make the mural seem ordinary and unremarkable—that is, unassuming.

Choice B is incorrect because in this context, "shrewd" would mean clever or astute, and there's nothing in the text to suggest that a casual yet misleading description that only mentions the apparent simplicity of the mural's location and subject matter would make the mural seem particularly clever or astute. Though it's possible that the text's assertion that the mural is a "complex, dynamic meditation" demanding "serious attention" might convey the idea that it's a particularly clever or astute—that is, shrewd—artwork, the text wouldn't suggest that a casual description that fails to account for the work's complexity would likewise characterize the work as shrewd. *Choice C* is incorrect because in this context, "incongruous" would mean unsuitable for its surroundings or internally inconsistent. The "casual description" presented in the text indicates that the painting is housed in a coffee shop and depicts women drinking coffee, which would imply that the mural may in fact be well suited to its surroundings, not that it's ill suited to its location. Moreover, none of the details mentioned in this description would suggest that the painting contains internal inconsistencies of any kind. *Choice D* is incorrect because in this context, "pretentious" would mean expressive of exaggerated importance, which wouldn't fit the contrast established in the text between a "casual description" of the painting and the work's deeper themes. Though the text argues that the painting is worthy of "serious attention," which would suggest its importance, the text doesn't suggest that the casual description would by contrast imply that the painting's importance is exaggerated. Rather, by merely mentioning the mural's relatively ordinary location and subject matter without reference to this complexity, the casual description conveys that the painting is more unassuming and understated than it actually is, not that it expresses an exaggerated self-importance.

QUESTION 6

Choice B is the best answer because it most accurately describes how the underlined sentence functions in the text as a whole. The text establishes that Jam hears Bitter's studio door open early in the morning, when it is still dark. Jam then listens to her mother's footsteps, which seem to create a song that runs through the floorboards. The text indicates that the sound of Bitter's footsteps convey to Jam that Bitter is done painting. The underlined sentence then builds on this idea, stating explicitly that Bitter's feet sing the news of her completion of a painting. Thus, the sentence stating that "Bitter's feet were singing the news" further emphasizes that Bitter's footsteps themselves reveal information to Jam.

Choice A is incorrect because the text provides no evidence that Jam prefers music to art. While the underlined sentence uses musical imagery to describe the vibrations Jam feels, this is figurative language that is describing the footsteps' vibrations; it isn't an indication of Jam's personal interests. **Choice C** is incorrect because the text doesn't suggest that Bitter herself sings while working on paintings. The references to singing in the text are figurative descriptions of how Bitter's footsteps sound or feel to Jam, not literal statements about Bitter singing. **Choice D** is incorrect because the underlined sentence only describes Jam's perception of her mother's footsteps. It doesn't provide any details about Aloe or his reaction to the painting; in fact, the text doesn't mention Aloe seeing the painting at all.

QUESTION 7

Choice B is the best answer because it most accurately describes the main purpose of the text, which is to showcase how Adam's flexibility and consideration strengthen his relationship with his great-uncle Richard. The text begins by stating that the meeting between the two characters is proceeding well and then describes Adam's thoughtful behavior: noticing his great-uncle's interest in the ship, Adam suggests staying on board overnight and expresses willingness to accommodate his great-uncle's schedule and destination preferences. The text goes on to describe the positive effect of Adam's thoughtfulness and flexibility, indicating that Adam's "affectionate willingness" to accommodate Richard's preferences "quite won the old man's heart." The text concludes by suggesting that Adam's ability to anticipate his great-uncle's wishes and willingness to accommodate them result in the rapid development of affection and intimacy between them, stating that they soon became "almost like old friends." Therefore, the text's main purpose is to demonstrate how Adam's flexibility and consideration strengthened his and his great-uncle's relationship.

Choice A is incorrect. Although the text explains Adam's reason for suggesting that Richard spend the night on the ship rather than finding lodging on land (which is that Adam perceives Richard's interest in the ship), this isn't the text's main purpose. Rather, Adam's suggestion about where to sleep for the night serves as an example of the consideration Adam shows toward Richard, and the text mentions it mainly to explain how an affectionate relationship developed so quickly between them. Moreover, the text only states that Adam encourages Richard to stay on the ship; it doesn't specify where Adam plans to spend the night. **Choice C** is incorrect because the text doesn't describe why Adam and his

great-uncle Richard are excited about their journey. While the text mentions that Adam is willing to “start at any hour and go anywhere,” it provides no details about any planned journey, its destination, or even whether they will be continuing by ship or some other mode of transportation, nor does the text mention anything about either character’s feelings of excitement about their plans. We learn only that Adam is accommodating about potential travel plans, not that either character is particularly eager about the journey itself. The text’s main focus is on Adam and Richard’s developing relationship, not on any anticipated travel experiences. *Choice D* is incorrect because the text doesn’t show a contrast between initial wariness and later affection. On the contrary, the text begins by stating that the meeting between Adam and his great-uncle Richard has begun in an “auspicious,” or promising, manner and is “proceed[ing] well,” suggesting that Richard’s impression is positive from the outset. There’s no indication of any initial suspicion, caution, or reserve on Richard’s part that later changes to affection. Instead, the text describes a consistently positive interaction that deepens from a good first impression into a stronger, more familiar relationship because of Adam’s considerate behavior.

QUESTION 8

Choice D is the best answer because it most accurately describes how the underlined portion functions in the text as a whole. The text first establishes that in order to investigate visual perception of objects in dreams, LaBerge and team observed the eye movements of lucid dreamers (people who are aware of their dreaming as it occurs) when these participants signaled that they were in a dream state. The underlined portion then describes a finding that rests on a comparison: the way participants’ eyes tracked objects in dream states was very similar to how sighted people track real objects when awake. Finally, the text states that the team concluded from this finding that objects in dreams are perceived as more than just products of imagination—implying that the team thinks the eye movements are an indication that the brain treats dream objects as more real than imaginary. Thus, the function of the underlined portion is to identify a circumstance that is comparable to one in the study and that helps justify LaBerge and team’s conclusion, presenting the close similarity of visual tracking in wakeful and dreaming scenarios as support for the implicit idea that the brain perceives dream objects not as imaginary but as somewhat or fully real objects.

Choice A is incorrect because the text never presents LaBerge and team’s initial hypothesis, only what question they were interested in studying and what they found. So although the underlined portion does present evidence, that evidence doesn’t serve to undermine any hypothesis in the text. *Choice B* is incorrect because the text gives no indication that LaBerge and team changed the focus of their study at any point or that the finding described in the underlined portion (or any other finding) was unexpected or surprising. *Choice C* is incorrect. Rather than illustrating an implication of the main finding by LaBerge and team, the underlined portion only presents the finding itself, stating that the team found that the movements of lucid dreamers’ eyes while dreaming were very similar to those of sighted people visually tracking objects while awake. The next part of the text indicates what the finding implied to the team but still doesn’t offer a real-world example as illustration.

QUESTION 9

Choice D is the best answer because the critics mentioned in Text 2 would argue that negative reactions to controversial public art are not solely the result of the nature of the art itself. The claim in Text 1 suggests that public artworks tend to provoke backlash when they are not “aesthetically and conceptually bland.” This implies that the content of public art alone can cause backlash. However, the critics in Text 2 offer a more nuanced explanation. They argue that norm-defying pieces of public art that are not “effectively integrated within their surroundings” are more likely to be judged harshly by the public. They also imply that a museum provides a context that prepares viewers to expect challenging art, while public environments that serve other purposes often do not. This indicates that the critics in Text 2 would contend that factors beyond the attributes of the art, like environment and presentation, play a significant role in how public art is received.

Choice A is incorrect because the critics in Text 2 do not argue that unconventionality is unrelated to public disagreement about a public artwork’s merits. Rather, they acknowledge that norm-defying pieces do face unfavorable reactions, but they attribute such reactions to contextual factors like poor integration with the environment rather than dismissing the role of unconventionality entirely. *Choice B* is incorrect because the critics in Text 2 do not advocate for restricting public art to universally appealing works. Instead, they argue that better integration with the surroundings and situating the work in appropriate contexts could help norm-defying art be better received by passersby. *Choice C* is incorrect because the critics in Text 2 do not address whether civic leaders and community members are easily satisfied by norm-reinforcing art. The critics in Text 2 instead argue that installation and integration with the surrounding environment factor into how norm-defying art is received by the general public, rather than disputing how people respond to conventional art.

QUESTION 10

Choice C is the best answer because it presents a conclusion about increasing entrepreneurship in early adulthood that can be reasonably inferred from the text. The text explains that Andrés Hincapié investigated why young adults are less likely than older adults to start businesses and found that lack of knowledge of practical aspects of starting a business is a barrier. The text then states that Hincapié found that providing relevant informational resources greatly helps address that problem. Offering young adults practical information about how to start a business would be an example of providing relevant informational resources, so Hincapié would most likely say it is a promising way to increase early adult entrepreneurship.

Choice A is incorrect because creating social networks of young adults interested (but presumably inexperienced) in starting a business wouldn’t involve a clear source of practical information for those young adults, and the text indicates that Hincapié identified offering relevant informational resources about starting a business as a way to significantly reduce the problem of young adults’ low participation in entrepreneurship. *Choice B* is incorrect because encouraging young people to brainstorm business ideas wouldn’t involve a clear source of

practical information for those young adults, and the text indicates that Hincapié identified offering relevant informational resources about starting a business as a way to significantly reduce the problem of young adults' low participation in entrepreneurship. *Choice D* is incorrect because providing young adults with training opportunities at existing businesses would give those young people exposure to businesses that already exist but not necessarily information about how to start a business, and the text indicates that Hincapié identified offering relevant informational resources about starting a business as a way to significantly reduce the problem of young adults' low participation in entrepreneurship.

QUESTION 11

Choice B is the best answer because it most accurately states the main topic of the text. The text begins by noting that few animals are known to spit, and then it focuses on Clare Jacobs's January 2022 observation of a gray seal spitting water at a white-tailed eagle. The text explicitly notes that this behavior had never before been observed in seals, highlighting that this was a previously unseen behavior. The text concludes with biologist Sean Twiss's hypotheses about why the seal might have been spitting. Thus, the main topic is the observation of a previously unseen behavior (spitting) of gray seals.

Choice A is incorrect because while the text does mention bird-watcher Clare Jacobs making an observation in southern England, bird-watching itself is not the main topic. The reference to bird-watching is incidental to the central focus of the novel behavior of a gray seal spitting water at an eagle. *Choice C* is incorrect because the text only mentions a white-tailed eagle as the target of a seal's spitting behavior; it doesn't provide any information about how these eagles typically behave when defending territory or interacting with other predators. *Choice D* is incorrect because the text doesn't discuss differences between gray seals and white-tailed eagles. While both animals are mentioned, the text is focused on the seal's unusual spitting behavior rather than comparing the two animals in any way.

QUESTION 12

Choice D is the best answer because it most accurately describes what is notable about Schmidt's behavior. According to the text, when the narrator's first wife admits she doesn't find art particularly compelling, Schmidt responds by wincing, sighing dramatically, and abruptly leaving dinner with an obviously false excuse about a remembered appointment. The narrator makes it clear that there is no actual appointment and that the excuse is merely a pretense. Through these actions—the physical reactions of wincing and sighing, followed by the transparent excuse to leave—Schmidt clearly communicates his disapproval of the wife without directly stating his feelings about her.

Choice A is incorrect because the text provides no evidence that Schmidt's theatrical behavior—wincing, sighing dramatically, and abruptly leaving dinner—is unique to situations involving the narrator and his first wife. The text describes only one interaction involving Schmidt, so there's no basis for determining

whether his dramatic reactions are typical of him in all situations or limited to this particular social context. *Choice B* is incorrect because nothing in the text suggests that Schmidt is typically mindful of his schedule or that forgetting appointments would be uncharacteristic of him. Rather, the text makes it clear that Schmidt hasn't actually forgotten an appointment at all—the narrator indicates that Schmidt's reason for leaving is an obvious fabrication, not a genuine instance of absentmindedness. *Choice C* is incorrect because the text doesn't mention any previous disagreement between Schmidt and the narrator's first wife about a particular painting or any other topic. The text presents what appears to be a spontaneous reaction by Schmidt to the wife's statement about not finding art compelling, not the continuation of an earlier conversation or dispute.

QUESTION 13

Choice A is the best answer because it accurately describes Sir Winston Day as he is portrayed in the text. The text indicates that Sir Winston's "features seemed moulded to adorn a bust, or possibly a stamp," suggesting that he appeared distinguished. But the text also indicates that his "actual achievements" during his long public service constitute an "enigma," or mystery, highlighting the point with a joke about Sir Winston's uninteresting memoirs (any information about his achievements past the first thirty pages "might as well have remained state secrets" since, it's implied, no one bothered to read that far). Thus, the text suggests that Sir Winston appears distinguished even though his known accomplishments, if any, are relatively few.

Choice B is incorrect because nothing in the text indicates that Sir Winston behaved dishonorably. Rather, the text indicates that his "actual achievements" during his long public service constitute an enigma, or mystery, suggesting that he may have done very little, not that what he has done was disreputable.

Choice C is incorrect. Although the text says that Sir Winston's achievements that appear "after the thirty-page mark" in his memoirs "might as well have remained state secrets," this is a joke about the public's lack of interest in the memoirs, not a sincere claim that secrecy was intended. *Choice D* is incorrect because the text does not associate Sir Winston with modesty, nor does it suggest that he was a capable public servant. Rather, by indicating that Sir Winston's "features seemed moulded to adorn a bust, or possibly a stamp," the text suggests that Sir Winston's appearance communicated importance, not modesty. The text further indicates that his "actual achievements" during his long public service constitute an enigma, or mystery, suggesting that he may have done very little, not that he was particularly capable.

QUESTION 14

Choice D is the best answer because it most accurately states the main idea of the text. The text begins by explaining that most studies of ocean wave breaking have focused on traveling waves (those that move horizontally) but that McAllister et al. studied spike waves, which form when traveling waves moving in opposing directions intersect. The text establishes that while traveling waves break when

their steepness exceeds a critical threshold (limiting their height), McAllister et al. found that spike waves can exceed these constraints because factors like jet stability and cavity shape can mitigate the effects of steepness. Thus, the main idea of the text is that spike waves resulting from traveling waves intersecting in specific ways can reach heights greater than what would be expected based on the general properties of traveling waves.

Choice A is incorrect because it misrepresents the information in the text; the text indicates that McAllister et al. showed that jet stability and cavity shape (along with steepness) influence breaking in spike waves, not in traveling waves. Further, the information about specific influences is not the main focus of the text.

Choice B is incorrect because it misrepresents the information in the text; the text indicates that McAllister et al. found that spike waves can pass the steepness threshold at which traveling waves break because factors like jet stability and cavity shape affect the breaking of spike waves, not because they've simply reached a critical threshold of steepness. *Choice C* is incorrect. Although the text does explain that spike waves can form when traveling waves intersect, it presents this as a known fact that McAllister et al.'s study was based on, not as something the researchers suggested. Further, the text indicates that spike waves can exceed a threshold that limits traveling wave height but doesn't focus on the idea that spike waves typically are higher than traveling waves.

QUESTION 15

Choice B is the best answer because it most effectively uses data from the graph to complete the sentence about which fuel has the highest energy density. Of the four bars in the graph, the highest (indicating the greatest energy density in MJ/L) is for POP biofuel. According to the graph, the energy densities of the fuels shown are approximately 40 MJ/L for POP biofuel, 36 MJ/L for gasoline, 34 MJ/L for jet fuel, and 23 MJ/L for ethanol.

Choice A is incorrect. The task is to find the fuel in the graph with the highest bar (indicating the greatest energy density in MJ/L), and jet fuel has the third-highest bar in the graph, not the highest. According to the graph, the energy densities of the fuels shown are approximately 40 MJ/L for POP biofuel, 36 MJ/L for gasoline, 34 MJ/L for jet fuel, and 23 MJ/L for ethanol. *Choice C* is incorrect. The task is to find the fuel in the graph with the highest bar (indicating the greatest energy density in MJ/L), and ethanol has the lowest bar in the graph, not the highest. According to the graph, the energy densities of the fuels shown are approximately 40 MJ/L for POP biofuel, 36 MJ/L for gasoline, 34 MJ/L for jet fuel, and 23 MJ/L for ethanol. *Choice D* is incorrect. The task is to find the fuel in the graph with the highest bar (indicating the greatest energy density in MJ/L), and gasoline has the second-highest bar in the graph, not the highest. According to the graph, the energy densities of the fuels shown are approximately 40 MJ/L for POP biofuel, 36 MJ/L for gasoline, 34 MJ/L for jet fuel, and 23 MJ/L for ethanol.

QUESTION 16

Choice A is the best answer because it most effectively illustrates the theater historian's claim that El Teatro Campesino's early performances were influenced stylistically by *carpa* theater, vaudeville-style shows with "audience-mediated slapstick comedy." The quotation indicates that El Teatro presented comedic sketches that often featured exaggerated physical humor and that the performers improvised based on the cheers and boos from the audience; thus, it illustrates that El Teatro's performances were similar to *carpa* theater in the physical, slapstick nature of their comedy and in the way the performances were mediated, or shaped, by the audience's reactions.

Choice B is incorrect. Although the quotation refers to El Teatro's use of comedy and mentions audiences, it focuses on the company's use of comedic aspects of reality as a tool to critique society while providing audiences with entertainment and inspiration; it doesn't give any indication that El Teatro's performances featured slapstick, or very physical, comedy or that they were directly shaped by the audience. **Choice C** is incorrect. Although the quotation refers to El Teatro's aesthetic, or style, and mentions comedy, it indicates only that the company relied on satire and humor and had a style that reflected their limited financial means and sociopolitical message; it doesn't give any indication that El Teatro's performances featured slapstick, or very physical, comedy or that they were directly shaped by the audience. **Choice D** is incorrect. Although the quotation refers to El Teatro's performance of comedy, it mentions only that the company performed brief humorous vignettes, or short humorous scenes, in farm fields; it doesn't give any indication that El Teatro's performances featured slapstick, or very physical, comedy or that they were directly shaped by the audience.

QUESTION 17

Choice D is the best answer because it most logically completes the text's discussion of the relationship between street lighting and people's cycling behavior. According to the text, Eugeni Vidal-Tortosa and Robin Lovelace found that poor street lighting deters new or inexperienced cyclists from riding their bikes in a city but has little effect on experienced cyclists. This finding establishes a causal relationship between street lighting and the willingness of new or inexperienced cyclists to ride a bicycle in the city: if poor lighting discourages new or inexperienced cyclists from riding, then improving lighting conditions by increasing the number of streetlights would logically remove this deterrent, potentially leading to an increase in the number of new or inexperienced cyclists willing to ride in the city.

Choice A is incorrect because it contradicts the logical relationship established in the text. If poor street lighting deters new or inexperienced cyclists from riding in the city, then increasing the number of streetlights (improving lighting conditions) would be expected to encourage more of these cyclists to ride, not decrease their numbers. **Choice B** is incorrect because the text explicitly states that street lighting didn't have an effect on experienced cyclists. If lighting conditions don't significantly influence experienced cyclists' decisions to ride, then increasing the number of streetlights wouldn't necessarily increase the number of experienced

cyclists riding in the city. *Choice C* is incorrect. The text indicates that street lighting didn't affect experienced cyclists, so increasing the number of streetlights wouldn't decrease the participation of these cyclists. Instead, their numbers would be expected to remain about the same.

QUESTION 18

Choice D is the best answer because it most logically completes the text's discussion of wild-bee declines. The text establishes that bumblebees and other wild bees have been experiencing population collapse as a result of habitat destruction, climate variation, and other factors. The text then indicates that because bumblebees are very extensively researched, ecologists rely heavily on findings about them to understand wild-bee declines in general. However, the text then introduces Ghisbain's observation that unlike most other wild-bee genera, bumblebees have certain traits (social behaviors and dietary generalism) linked to increased resilience to certain environmental changes. In other words, bumblebees aren't necessarily representative of wild bees as a whole because they are likely more tolerant of some pressures. Therefore, it logically follows that Ghisbain would urge researchers to exercise caution when using bumblebee data to draw conclusions about other wild-bee population declines, since bumblebees and other wild bees don't always respond comparably to environmental threats.

Choice A is incorrect because the text doesn't indicate that climate variation is more of a threat to bumblebees than habitat destruction is or that it is a bigger threat to bumblebees than it is to other wild bees. The text simply indicates that climate variation and habitat destruction are among the factors that have caused population collapses for bumblebees and other wild-bee genera. *Choice B* is incorrect. As it is presented in the text, Ghisbain notes that bumblebees may respond to environmental changes differently than many other wild-bee genera do; this suggests that bumblebees shouldn't be treated as representative of all wild bees, but there's no reason to think that Ghisbain would go so far as to assert that bumblebee data shouldn't be used to draw conclusions about other wild-bee genera that are similar to bumblebees in their dietary and social traits, even if there are relatively few. The text suggests caution in extrapolation, not a complete rejection of using bumblebee research. *Choice C* is incorrect. Although the text indicates that dietary generalism is linked to increased resilience to specific environmental changes (that is, to some but not all environmental stresses), it doesn't suggest that bumblebees and other bees with this trait are less likely than bees with specialized diets to experience major population changes. In fact, the text makes it clear that like many other wild-bee genera—most of which don't display dietary generalism—bumblebees are already experiencing population collapse.

QUESTION 19

Choice D is the best answer. The convention being tested is end-of-sentence punctuation. This choice correctly uses a period to punctuate a declarative sentence that ends with an indirect question ("which link you should choose.")

Choice A is incorrect. The structure requires a declarative clause that presents an indirect question about what is evaluated in click restraint, not an interrogative clause that presents a direct question, such as “which link should you choose.” *Choice B* is incorrect. It’s unconventional to use a question mark in this way to punctuate a declarative clause that presents an indirect question, such as “which link you should choose.” *Choice C* is incorrect. The structure requires a declarative clause that presents an indirect question about what is evaluated in click restraint, not an interrogative clause that presents a direct question, such as “which link should you choose.”

QUESTION 20

Choice D is the best answer. The convention being tested is the use of verbs to express tense in a sentence. The past perfect tense indicates an action that was completed before another past action. In this choice, the past perfect verb “had become,” together with the supplemental phrase “by 2023,” indicates that Lafourcade earned her status as one of the most celebrated musicians in Latin America before 2023.

Choice A is incorrect because the future tense “will become” doesn’t indicate that Lafourcade earned her celebrated status before 2023. *Choice B* is incorrect because the present tense “becomes” doesn’t indicate that Lafourcade earned her celebrated status before 2023. *Choice C* is incorrect because the future perfect tense “will have become” doesn’t indicate that Lafourcade earned her celebrated status by 2023.

QUESTION 21

Choice A is the best answer. The convention being tested is the use of verbs to express tense in a sentence. In this choice, the present tense verb “includes,” used in conjunction with the word “currently,” correctly indicates that the frog’s current range includes parts of northern New South Wales and southeastern Queensland.

Choice B is incorrect because the past tense verb “included” doesn’t indicate that the frog’s range currently includes parts of northern New South Wales and southeastern Queensland. *Choice C* is incorrect because the future perfect tense “will have included” doesn’t indicate that the frog’s range currently includes parts of northern New South Wales and southeastern Queensland. *Choice D* is incorrect because the past perfect tense “had included” doesn’t indicate that the frog’s range currently includes parts of northern New South Wales and southeastern Queensland.

QUESTION 22

Choice B is the best answer. The convention being tested is the use of punctuation within a sentence. The comma after “1804” pairs with the comma after “amendment” to separate the supplementary element “ratified in 1804” from the rest of the sentence. This supplementary element functions to provide additional information on the term “12th amendment,” and the pair of commas indicates that this element could be removed without affecting the grammatical coherence of the sentence.

Choice A is incorrect because a dash can't be paired with a comma in this way to separate the supplementary element from the rest of the sentence. *Choice C* is incorrect because a colon can't be paired with a comma in this way to separate the supplementary element from the rest of the sentence. *Choice D* is incorrect because it fails to use appropriate punctuation to separate the supplemental element from the rest of the sentence.

QUESTION 23

Choice B is the best answer. The convention being tested is subject-modifier placement. This choice makes the noun phrase “the titles” the subject of the sentence and places it immediately after the modifying phrases “laden with emojis and seemingly meaningless words, and driven largely by Sawano’s ‘personal feeling and mood.’” In doing so, this choice clearly establishes that the titles—and not another noun in the sentence—are being described as laden with emojis and driven by Sawano’s personal feeling.

Choice A is incorrect because it results in a dangling modifier. The placement of “the listener” immediately after the modifying phrases illogically suggests that the listener, rather than the titles, is laden with emojis and driven by Sawano’s personal feeling. *Choice C* is incorrect because it results in a dangling modifier. The placement of “each piece” (i.e., musical score) immediately after the modifying phrases illogically suggests that the pieces, rather than the titles of the pieces, are laden with emojis and driven by Sawano’s personal feeling. *Choice D* is incorrect because it results in a dangling modifier. The placement of “the listener’s approach” immediately after the modifying phrases illogically suggests that the listener’s approach, rather than the titles, is laden with emojis and driven by Sawano’s personal feeling.

QUESTION 24

Choice A is the best answer. The convention being tested is subject-verb agreement. The plural verb “explore” agrees in number with the plural subject “Barrada’s pieces.”

Choice B is incorrect because the singular verb phrase “has explored” doesn’t agree in number with the plural subject “Barrada’s pieces.” *Choice C* is incorrect because the singular verb “explores” doesn’t agree in number with the plural subject “Barrada’s pieces.” *Choice D* is incorrect because the nonfinite participle “exploring” doesn’t supply the main clause with a finite verb.

QUESTION 25

Choice D is the best answer. The convention being tested is the use of punctuation within a sentence. The comma after “swimming” pairs with the dash after “ciliates” to offset the supplementary element (“widely...ciliates”) that interrupts the main clause of the sentence.

Choice A is incorrect. Adding the verb “is” here creates an ungrammatical sentence structure with two main verbs (“is” and “bestows”) without appropriate punctuation and/or conjunction. *Choice B* is incorrect because it fails to offset the

supplementary element with appropriate punctuation. Furthermore, adding the verb “is” here creates an ungrammatical sentence that has two finite verbs (“is” and “bestows”) without appropriate punctuation and/or conjunction. *Choice C* is incorrect because it fails to offset the supplementary element with appropriate punctuation. Furthermore, adding the verb “has been” here creates an ungrammatical sentence that has two finite verbs (“has been” and “bestows”) without appropriate punctuation and/or conjunction.

QUESTION 26

Choice D is the best answer. “As a result” logically signals that the information in this sentence—that the widened grooves in balsa wood can accommodate electrical conductors—is a direct consequence of the researchers’ method described in the previous sentence (using heat and chemicals to widen the grooves).

Choice A is incorrect because “for example” illogically signals that this sentence provides an example of the researchers’ method described in the previous sentence. Instead, the sentence indicates a consequence of that method. *Choice B* is incorrect because “previously” illogically signals that this sentence describes an event that occurred before the researchers’ development of the method described in the previous sentence. Instead, the sentence indicates a consequence of that method. *Choice C* is incorrect because “in contrast” illogically signals that the information in this sentence contrasts with the previous sentence’s description of the researchers’ method. Instead, the sentence indicates a consequence of that method.

QUESTION 27

Choice D is the best answer. “Consequently” logically signals that the information in this sentence—that many Arabic words entered the Spanish language—is a result or consequence of the previous information about Arabic-speaking Muslim states ruling Al-Andalus for centuries.

Choice A is incorrect because “for example” illogically signals that the information about Arabic words entering Spanish exemplifies the Arabic-speaking history of Al-Andalus mentioned in the previous sentence. Instead, Arabic words entering Spanish is a consequence of that history. *Choice B* is incorrect because “instead” illogically signals that the information about Arabic words entering Spanish provides an alternative to the Arabic-speaking history of Al-Andalus mentioned in the previous sentence. The information about Arabic words entering Spanish is a consequence of that history. *Choice C* is incorrect because “specifically” illogically signals that the information about Arabic words entering Spanish provides specific details elaborating on the Arabic-speaking history of Al-Andalus mentioned in the previous sentence. Instead, Arabic words entering Spanish is a consequence of that history.

QUESTION 28

Choice A is the best answer. “By contrast” logically signals that the information in this sentence—that paper scraps left over from printing a magazine are pre-consumer waste—contrasts with the previous information about magazines being post-consumer waste.

Choice B is incorrect because “for example” illogically signals that the information in this sentence exemplifies the previous information about post-consumer waste. Instead, the paper scraps being pre-consumer waste contrasts with the previous information. *Choice C* is incorrect because “as a result” illogically signals that the information in this sentence is a result or consequence of the previous information about post-consumer waste. Instead, the paper scraps being pre-consumer waste contrasts with the previous information. *Choice D* is incorrect because “specifically” illogically signals that the information in this sentence provides specific, precise details elaborating on the previous information about post-consumer waste. Instead, the paper scraps being pre-consumer waste contrasts with the previous information.

QUESTION 29

Choice A is the best answer. “Though” logically signals that the information in this sentence—that Morgado and colleagues detected a dense ring orbiting outside the Roche limit that remained intact—presents an exception to the previous sentence’s claim about debris orbiting outside an object’s Roche limit. Such debris would be expected to accrete into a satellite, not remain intact as a ring.

Choice B is incorrect because “for example” illogically signals that the information about Morgado’s discovery exemplifies the previous sentence’s claim about debris orbiting outside an object’s Roche limit. Instead, the sentence presents an exception to the claim. *Choice C* is incorrect because “fittingly” illogically signals that the information about Morgado’s discovery aligns with the previous sentence’s claim about debris orbiting outside an object’s Roche limit. Instead, the sentence presents an exception to the claim. *Choice D* is incorrect because “likewise” illogically signals that the information about Morgado’s discovery is similar to the previous sentence’s claim about debris orbiting outside an object’s Roche limit. Instead, the sentence presents an exception to the claim.

QUESTION 30

Choice C is the best answer. The sentence indicates how long John Cage’s musical piece will last, noting that it will last 639 years and be the longest continuous musical performance in history.

Choice A is incorrect. The sentence identifies John Cage as the composer and provides the name of the musical piece; it doesn’t indicate how long the piece will last. *Choice B* is incorrect. The sentence provides the name of the musical piece and where it is currently being played; it doesn’t indicate how long the piece will last. *Choice D* is incorrect. The sentence mentions where the musical piece is being played and identifies John Cage as the composer; it doesn’t indicate how long the piece will last.

QUESTION 31

Choice A is the best answer. The sentence provides a historical overview of the Shackleton expedition, noting that the explorers left for Antarctica in 1914, lost their ship in 1915, and were rescued in 1916.

Choice B is incorrect. The sentence misrepresents information from the notes; Shackleton and his crew were themselves rescued in 1916—they weren't rescuing others. **Choice C** is incorrect. The sentence provides examples of works and discoveries inspired by the expedition; it doesn't provide an overview of the expedition itself. **Choice D** is incorrect. The sentence provides examples of works made about the expedition; it doesn't provide an overview of the expedition itself.

QUESTION 32

Choice D is the best answer. Noting that *Meghadūta* uses a pattern of long and short syllables in its lines (quantitative meter) and *Widsith* uses a pattern of repeated sounds in its lines (alliterative meter), and signaling a contrast with “whereas,” the sentence emphasizes a difference between the meters of the two poems.

Choice A is incorrect. The sentence indicates that the two poems were written in different languages; it doesn't emphasize a difference between the meters of the two poems. **Choice B** is incorrect. The sentence mentions the meters of the two poems but misrepresents information from the notes; the overall definition of a meter (the rhythmic structure or pattern of accents in a poem's lines) applies to both alliterative and quantitative meters, not just quantitative. **Choice C** is incorrect. While the sentence emphasizes a difference between the meters of the two poems, it misrepresents information from the notes; *Widsith* uses an alliterative meter, which is structured by a pattern of repeated sounds, not a pattern of long and short syllables.

QUESTION 33

Choice B is the best answer. The sentence places the Declaration of Independence in the context of Valle's changing political beliefs, noting that Valle was long an opponent of Central American independence but changed his mind after an 1820 revolt and then wrote the Declaration of Independence in 1821.

Choice A is incorrect. The sentence states that Valle's political beliefs changed after Riego's revolt but doesn't indicate how these changed beliefs provide context for the 1821 Declaration of Independence. **Choice C** is incorrect because it misrepresents information from the notes. Valle's political beliefs changed after Riego's 1820 revolt, not when the Spanish king repealed the 1812 constitution. **Choice D** is incorrect. While the sentence indicates a relationship between Colonel Riego's 1820 revolt and Central America's Declaration of Independence, it doesn't mention Valle's changing political beliefs.

Reading and Writing

Module 2

(33 questions)

QUESTION 1

Choice C is the best answer because it most logically completes the text's discussion of the ancient Inca farming method. In this context, "effective" means successful in producing a desired result. The text describes how the Inca used terraces with built-in irrigation systems to grow crops at different altitudes and then indicates that this farming method produced positive results "as evidenced by the great number and variety of crops grown at that time." The context clearly suggests that the terraced farming approach was highly productive, or effective.

Choice A is incorrect because to describe the Inca farming method as "chaotic" contradicts the text's description of a systematic approach using deliberately constructed terraces with built-in irrigation systems. Moreover, the "great number and variety of crops" yielded by this method seems unlikely to result from a disorganized, or chaotic, approach. *Choice B* is incorrect because "uniform" means consistent or the same throughout. It does not make sense in this context to describe a farming method as highly uniform if it produced a "great number and variety of crops." Moreover, the text indicates that Inca terrace farming was specifically designed to accommodate different crops at different altitudes, highlighting the diversity rather than uniformity of this approach. *Choice D* is incorrect because the success of the method, "as evidenced by the great number and variety of crops grown," does not support the claim that the Inca farming method was "burdensome," or difficult to implement or maintain. In fact, the context clearly emphasizes the advantages, rather than the burdens, of Inca terrace farming.

QUESTION 2

Choice C is the best answer because it most logically completes the text's discussion of repairing the Hubble Space Telescope. In this context, "relying on" means depending on. The text states that astronauts have regularly had to go on missions to repair the Hubble Space Telescope but that robots might soon be able to make the repairs, which would be helpful because involving astronauts in

the process is expensive. It is most reasonable to assume that it is the need to send astronauts on missions that is expensive; thus, the context suggests that employing robots for repairs will be helpful because depending on repeatedly sending humans to space is costly.

Choice A is incorrect. In this context, “straightening” would mean physically aligning or removing curves and bends. It’s not clear how astronauts would be physically straightened to maintain the telescope or why that would be expensive. *Choice B* is incorrect because the text suggests that it is the ongoing involvement of astronauts—the need to send them to space regularly—that is expensive. Nothing in the text indicates that astronauts have been forgotten about or that forgetting about them would be expensive. *Choice D* is incorrect. In this context, “reducing” would most likely mean lowering in status, and nothing in the text suggests that astronauts are diminished in any way by repairing the telescope (indeed, they have regularly done repairs for years). Further, it isn’t clear how a decrease in status would be expensive.

QUESTION 3

Choice C is the best answer because it most logically completes the text’s discussion of election polling. In this context, to “neglect” would mean to disregard or pay insufficient attention to something. The text states that there have been many cases of election pollsters incorrectly predicting presidential elections and then says that neuroscientist and pollster Sam Wang notes that campaigns shouldn’t react to these failures in a certain way. The text then adds that polls are valuable for informing strategists’ efforts throughout campaigns, not just for predicting the outcome. This context conveys that Wang’s advice is not to neglect, or disregard, polling because even though it sometimes fails to predict the winner, it can help improve campaign strategies.

Choice A is incorrect. In this context, to “distort” would mean to give a false or misleading account. Though the text states that many election pollsters have failed in their predictions in presidential elections, it gives no indication that Wang would therefore argue that campaigns shouldn’t react to such polling failures by giving inaccurate accounts of election polling. Instead, the text emphasizes that polls have value beyond predicting winners, suggesting that Wang’s point is that campaigns shouldn’t stop polling entirely. *Choice B* is incorrect because it wouldn’t make sense to suggest that Wang encourages campaigns not to “enact,” or establish, election polling; the text indicates that election polling is something that already takes place and that has value for campaign strategies even if it sometimes leads to inaccurate predictions of winners, suggesting an argument in favor of continuing to use election polling. *Choice D* is incorrect. Although the text indicates that many election pollsters have failed in their predictions in presidential elections, there’s no reason to think Wang would say that these failures should lead campaigns *not* to “supplement,” or add to, election polling; it would be reasonable to suggest that campaigns *would* want to consider more than just election polling information if that information may lead to inaccurate predictions. Moreover, the text’s main point is that polls provide some utility even if their predictions may be inaccurate, which suggests that Wang’s recommendations pertain to whether campaigns should or shouldn’t use polling data, not whether polling data should or shouldn’t be supplemented with other kinds of information.

QUESTION 4

Choice D is the best answer because it most logically completes the text's discussion of Steiger Butte Drum's contributions to the musical work *Natural History*. In this context, "integral" means essential or necessary to the whole. The text states that Steiger Butte Drum not only took part in the piece's creation, collaborating with composer Michael Gordon on it, but also "must be included in all performances." This context clearly indicates that the ensemble's participation is essential, or integral, to the piece, both in its creation and its performance.

Choice A is incorrect because in this context, "tangential" would mean peripheral or of little relevance to something. The text emphasizes the importance of Steiger Butte Drum's involvement in both the composition and performance of *Natural History*, indicating their role is central, not tangential. **Choice B** is incorrect because in this context, "subsequent" would mean following after or coming later than something else. The text doesn't establish a temporal sequence where the participation of Steiger Butte Drum followed after the creation of *Natural History*; rather, it indicates that the ensemble was involved with the piece from the beginning, having worked with Michael Gordon during the creation process. **Choice C** is incorrect because in this context, "analogous" would mean similar or comparable to something else, and the text doesn't compare Steiger Butte Drum's participation in the creation and performances of *Natural History* to anything else; instead, it describes the nature and importance of the ensemble's involvement.

QUESTION 5

Choice B is the best answer because as used in the text, "manifest" most nearly means perceptible or detectable. The text indicates that the sun was "already down," meaning it was below the horizon and not directly visible. However, the sun's presence was nonetheless perceptible or manifest in the "tongues of fire" (or beams of light) that still flickered in the darkening sky.

Choice A is incorrect. Although in some contexts "manifest" can mean "realized" or "brought into being," that is not the context here. Rather, the text is about how the sun's presence was perceptible or manifest in the "darting" beams of light that could still be seen in the sky. **Choice C** is incorrect. This choice would suggest that the sun is situated, or located, in the light it emits, which is not supported by the surrounding context. Rather, the text is about how the sun's presence was perceptible or manifest in the remaining light that could still be seen in the sky. **Choice D** is incorrect. Although the text is about the decreasing light in the scene at sunset, the text's use of "manifest" is about how the sun's presence was nonetheless perceptible or manifest in the "tongues of fire" still visible in the sky as night approached.

QUESTION 6

Choice B is the best answer because it most accurately describes how the underlined portion functions in the text as a whole. The underlined portion is a parenthetical phrase directly following the word "microfilm," and it describes film containing small ("scaled-down") images; the placement and focus of the portion indicate that it is serving to define the term "microfilm."

Choice A is incorrect. Although the text is about accessing the writings of a famous historical figure (Douglass), the underlined portion doesn't give information about Douglass or any other person. Instead, it explains what microfilms are. *Choice C* is incorrect because the text never describes any kind of debate or disagreement; it simply describes a change in how researchers can access historical documents, with the underlined portion defining an important term. *Choice D* is incorrect because the text never makes any mention of a particular finding, unexpected or otherwise; it simply describes a change in how researchers do some of their work, with the underlined portion defining an important term.

QUESTION 7

Choice B is the best answer because it most accurately describes the overall structure of the text. The text begins by stating that the gray wolf was eradicated from Yellowstone National Park in 1926. It then explains the consequences of this elimination—namely that the elk population grew too large and overgrazing occurred, negatively affecting other animals in the area. The text then states that once scientists realized the importance of the gray wolf to the Yellowstone food chain, they undertook efforts leading to the wolves' reintroduction to the park in 1996. Thus, the structure of the text can be accurately described as mentioning the elimination of the gray wolf from Yellowstone and then explaining why the wolf was later restored to the park.

Choice A is incorrect. While the text does summarize a problem associated with an event in the 1920s (the eradication of gray wolves from Yellowstone in 1926), the problems discussed arose after that event—namely, the subsequent elk overpopulation, resultant overgrazing, and then further impacts to other animals in Yellowstone. Furthermore, the text doesn't present multiple potential solutions. Instead, it describes one solution that was actually implemented: the reintroduction of gray wolves in 1996. *Choice C* is incorrect because the text doesn't make a claim about the health of the gray wolf population apart from pointing out that the wolves had been eradicated from, and then subsequently restored to, Yellowstone. Instead, the text focuses on outcomes associated with the gray wolf's eradication and reintroduction. *Choice D* is incorrect. Though the text does mention that the ecosystem of Yellowstone rebounded after the gray wolf was reintroduced in 1996, the text never explains why the park allowed the wolves to be eradicated in 1926. It simply states that this eradication occurred without elaborating on the decision-making process behind it.

QUESTION 8

Choice A is the best answer because it most accurately describes the overall structure of the text. The text first presents what researchers found from studying Matabele ants: the ants can tell when colony members have infected wounds and can largely successfully treat those infections with their antimicrobial secretions. The text then indicates that the findings about the ants could help with the future development of antibiotics for humans. Thus, the text summarizes research findings and then identifies an area for further research.

Choice B is incorrect. Although the text introduces a study about Matabele ants, it never discusses how the researchers did their work or any methods they used to produce their findings. *Choice C* is incorrect. Although the text does describe properties of Matabele ants—their ability to detect and largely successfully treat infected wounds in colony members—it never addresses evolution or how these properties might have changed. *Choice D* is incorrect because the text doesn't present anything about Matabele ants as an issue or problem to be solved. Instead, it discusses beneficial abilities the ants have (detecting and treating infected wounds) and suggests that understanding those abilities might be useful for human medicine.

QUESTION 9

Choice B is the best answer because it most accurately describes the main purpose of the text, which is to explain the basics of how a specific energy technology works. The text introduces Raccoon Mountain as a pumped-storage hydropower facility built in the 1970s and then explains how it operates: during periods of low demand, excess power pumps water up to a lake at the mountain's summit, and when demand for energy peaks, the water flows back down, spinning turbines along the way to generate electricity. The text provides some additional details, like the name of the water source, but the overall focus is on explaining the basics of how pumped-storage hydropower works.

Choice A is incorrect because the text focuses on one method of energy generation—pumped-water hydropower—and briefly mentions another type of energy generation (nuclear plants) only to identify it as a source of power for the pumping process. The text never makes any comparison between the two methods. *Choice C* is incorrect because the text simply gives a factual description of how Raccoon Mountain, a pumped-storage hydropower facility, operates; it never suggests that more energy storage facilities should be built, whether by regional utilities or by any other entity. *Choice D* is incorrect because the text doesn't present pumped-storage hydropower as a new technology; it indicates that Raccoon Mountain was built in the 1970s, meaning that, rather than being new, this facility is approximately 50 years old.

QUESTION 10

Choice C is the best answer because it most accurately states the main idea of the text. The text begins by establishing Ruth Asawa as an artist who worked in several art forms but emphasizes that she was interested in art for reasons other than her desire to create it. The text mentions two ways in which Asawa brought art to children through public schools: cofounding the Alvarado School Arts Workshop in 1968 and helping found a public arts high school in 1982. Thus, the main idea is that Asawa's interest in art education motivated her to establish arts programs for students in San Francisco.

Choice A is incorrect because the text doesn't mention Asawa inspiring other artists to share their work with students. While the text does note that the Alvarado School Arts Workshop brought works of art and artists into public schools, it doesn't state that Asawa motivated these artists. *Choice B* is incorrect

because the text makes no comparison between local and national appreciation of Asawa's sculptures. While the text does mention that Asawa created unique tied-wire sculptures and was connected to San Francisco, it does not contrast her local and national critical receptions. *Choice D* is incorrect. Although the text emphasizes Asawa's educational initiatives, it doesn't state or imply that she abandoned her career as a sculptor to work as an art teacher. Instead, the text portrays her educational work as a complement to her career as a successful artist.

QUESTION 11

Choice B is the best answer because the graph shows that Montréal had the largest population in 1891 among the four cities represented. According to the data points in the graph for the year 1891, Montréal's population was approximately 220,000, which is greater than the respective populations of Toronto (approximately 180,000), Québec City (approximately 60,000), and Halifax (approximately 40,000).

Choice A is incorrect because Toronto's population in 1891 was approximately 180,000, which is the second largest among the four cities shown but still clearly smaller than Montréal's population of approximately 220,000. *Choice C* is incorrect because Québec City had a population of approximately 60,000 in 1891, making it only the third largest of the four cities shown. *Choice D* is incorrect because Halifax had a population of approximately 40,000 people in 1891. Of the four cities shown in the graph, Halifax has the smallest population, not the largest.

QUESTION 12

Choice D is the best answer because it accurately describes data from the graph that support the researchers' conclusion that children may be more aware of mammals than of other animals. The graph presents the percent of drawings by schoolchildren that contained an example of an animal from a certain group (mammals, birds, or insects). The graph shows that approximately 80% of the drawings contained an example of a mammal, while only approximately 69% contained an example of a bird and only about 55% contained an example of an insect. The fact that mammals were included in more drawings than either birds or insects supports the idea that children may be particularly aware of mammals in their environments.

Choice A is incorrect because the graph shows that there were not "about as many" drawings with insects as there were drawings with birds (approximately 69% of the drawings included an example of a bird, while only about 55% included an example of an insect). Further, comparing only the numbers for birds and insects wouldn't indicate anything about the schoolchildren's awareness of mammals. *Choice B* is incorrect because the graph shows that insects, not birds, were the group that appeared in the fewest number of drawings (approximately 69% of the drawings included an example of a bird but only about 55% included an example of an insect). Further, stating that examples from one nonmammal group appeared in the fewest number of drawings would provide no evidence about the schoolchildren's awareness of mammals. *Choice C* is incorrect because

the graph shows that more drawings had an example of a mammal than an example of an insect, not the other way around (approximately 80% of the drawings included an example of a mammal, while only about 55% included an example of an insect). Further, examples of a nonmammal group appearing in more drawings than examples of mammals did would actually weaken the claim that schoolchildren may be more aware of mammals than of other animals.

QUESTION 13

Choice C is the best answer because it accurately uses data from the graph to complete the statement regarding the research institute’s findings about veteran representation in Congress. The graph shows that the percentage of veterans in both houses of Congress decreased dramatically from 1973 to 2013. In 1973, the House of Representatives consisted of approximately 74% veterans, and the Senate was made up of approximately 79% veterans. By 2013, veteran members of both houses had fallen to approximately 17%–18%. This represents a substantial decrease of more than 50% in each house over this 40-year period.

Choice A is incorrect because from 2003 to 2023, the House of Representatives did not consistently have a higher percentage of veterans than the Senate did. According to the graph, in 2003 the Senate was made up of approximately 35% veterans, while the House consisted of approximately 23% veterans. In 2013, the number of veterans in the Senate was still slightly higher (approximately 18%) than in the House (approximately 17%). Only in 2023 did the House have a marginally higher percentage than the Senate. *Choice B* is incorrect because veterans did not constitute a majority in both houses throughout the entire period shown in the graph from 1953 to 2003. According to the graph, both houses had a majority of veterans in 1953 (approximately 62% in the House and approximately 74% in the Senate) and in 1973 (approximately 74% in the House and approximately 79% in the Senate), but by 1993 the House had fallen below 50% to approximately 40% veterans. By 2003, both houses were well below 50%.

Choice D is incorrect because the percentage of veterans in Congress did not remain consistent from 1993 to 2023. The graph shows significant changes in the number of members reporting past military service during this period, particularly the continued decline from 1993 to 2003. In 1993, the House had approximately 40% veterans and the Senate had approximately 63% veterans, but by 2003 these percentages had dropped to approximately 23% and 35%, respectively. Furthermore, there was a large gap between the two houses in 1993 that narrowed considerably by 2013, which contradicts the claim that the percentage of veterans across both houses “remained fairly consistent” during this period.

QUESTION 14

Choice B is the best answer because it most effectively illustrates the speaker of the poem’s claim that the stories Aunt Sue tells are based on Aunt Sue’s personal experiences. In the quotation, the speaker conveys the listening child’s point of view that Aunt Sue’s stories originate from her personal experiences and are not taken from books or other external sources. The listening child explicitly contrasts two possible sources for Aunt Sue’s stories: a book (whose potentially fictitious stories are implied to be written by someone else) or “her own life.” By having the

child confidently assert that Aunt Sue “never got her stories out of any book at all,” the speaker indicates that Aunt Sue’s stories are instead informed by her lived experiences.

Choice A is incorrect because it describes an element or image that recurs within Aunt Sue’s stories rather than indicating that these stories have a real-life basis. The reference to dark shadows that “cross and recross” the stories functions as imagery on a literal, visual level and may also evoke figurative darkness, but this detail by itself doesn’t suggest that the stories are based on Aunt Sue’s personal experiences. *Choice C* is incorrect because it identifies the setting of Aunt Sue’s storytelling rather than describing the real-life basis of the stories themselves. The quotation states that Aunt Sue tells her stories on “summer nights on the front porch,” which establishes when and where the storytelling occurs; however, no information is provided about the origins of the stories that Aunt Sue tells within this setting. *Choice D* is incorrect because it describes the movement or circulation of characters in the stories as Aunt Sue speaks rather than establishing that these stories have a real-life basis. The quotation primarily evokes the sounds of Aunt Sue’s voice and the seemingly easy “flow” of this voice in her storytelling. Although the characters who “mingle...in the flow of old Aunt Sue’s voice” are to some extent created or animated by her voice, no information is provided about the characters themselves or their relationship to Aunt Sue’s actual experiences.

QUESTION 15

Choice C is the best answer because it presents a finding that, if true, would most strongly support Manenti and team’s claim that olms regularly come to the surface to find food. The text explains that scientists previously believed olms remained in their underwater caves throughout their lives, but Manenti’s team claims that olms actually surface regularly to perform important activities like finding food. Since earthworms from surface soils would not be present in underwater caves, the finding that such earthworms constitute a major part of olms’ diet would provide compelling evidence that olms regularly leave their underwater cave environment to obtain food from the surface, thus supporting the underlined claim.

Choice A is incorrect because information about olms’ breeding frequency wouldn’t support the claim about their leaving their underwater habitats to obtain food. While breeding patterns might be relevant to understanding olm behavior in general, this finding doesn’t address whether olms surface to find food, which is the specific behavior that the underlined claim addresses. *Choice B* is incorrect because learning that olms live in only a few cave systems wouldn’t support the claim about their regular visits to the surface. This finding would provide information about the geographic distribution of olm habitats but wouldn’t address whether olms leave these caves to find food or perform other activities on the surface. *Choice D* is incorrect because the differences between olms’ brains and other salamanders’ brains have no clear connection to olms’ movement beyond their underwater habitats or to their feeding habits. A finding about the uniqueness of olms’ brains wouldn’t address whether olms leave their caves to find food or support Manenti and team’s claim about olms’ motivation for coming to the surface.

QUESTION 16

Choice C is the best answer because it most directly supports the text's claim that Stevens strove to convey the ephemeral and dynamic characteristics of *ikaah* as a cultural practice. The text challenges the characterization of Stevens's preserved sandpaintings as mere "static objects" that cannot "authentically represent" a practice defined in part by the absence of an end product; it claims instead that this "overly object-focused" interpretation overlooks how Stevens preserved the essential characteristics of traditional *ikaah* practice. This quotation supports the claim by explaining that Stevens's sandpaintings should be understood "not as self-contained objects but as reminders of the public creations of the sandpaintings," during which Stevens "conducted appropriate *ikaah* rituals and encouraged viewers to closely track his movements and subtle shifts in the sand throughout the process." The fact that Stevens created the sandpaintings in a public setting and invited viewers to attentively engage with this process supports the claim that he sought to portray *ikaah* as a dynamic practice rather than as a static product. Further, the inherently impermanent nature of such a performance evokes the ephemerality that characterizes *ikaah*.

Choice A is incorrect because it contradicts rather than supports the text's claim that Stevens strove to convey the ephemeral and dynamic characteristics of *ikaah* as a cultural practice. This quotation argues that Stevens's preserved sandpaintings "should not be confused with authentic *ikaah*" and emphasizes that authentic *ikaah* "cannot be extricated from a practice that is intentionally and necessarily transitory." This view aligns with the "overly object-focused" perspective that the text argues against, rather than supporting the claim that Stevens successfully conveyed *ikaah*'s ephemeral and dynamic characteristics.

Choice B is incorrect because it focuses on reconciling the tension between ephemeral and permanent forms rather than supporting the claim that Stevens actively conveyed *ikaah*'s ephemeral and dynamic characteristics. While this quotation acknowledges an apparent contradiction, it suggests that Stevens was accommodating US preferences for permanent artworks; it doesn't demonstrate how he communicated the ephemeral and dynamic nature of traditional *ikaah* practice. **Choice D** is incorrect. Although the quotation mentions Stevens's "temporary and performative role" as a gift-giver, it doesn't address how Stevens conveyed the ephemeral and dynamic characteristics of *ikaah* itself during his ambassadorial activities. The quotation focuses on Stevens's role in transmitting knowledge and representing cultural dynamism but doesn't specifically support the claim about his efforts to communicate the essential characteristics of *ikaah* practice. Further, the quotation incorrectly implies that Stevens shared his knowledge of preservative additives; the text doesn't establish that he revealed how the sandpaintings were preserved.

QUESTION 17

Choice A is the best answer because it most logically completes the text's discussion of mountain climbers' motivations for exaggerating the difficulty of reaching edelweiss. The text establishes that edelweiss became symbols of strength and courage in Switzerland in the mid-1800s. The text goes on to state that mountain climbing became popular around the same time and that climbers

spread the false idea that the flowers grew only in dangerous, hard-to-reach locations. According to the text, historian Tobias Scheidegger argues that these claims were “self-interested,” meaning they served the climbers’ own purposes. The text therefore implies that by claiming that edelweiss could be found only in perilous locations, mountain climbers could enhance their own reputations: climbers who encountered edelweiss would appear to have undertaken a dangerous journey, thereby making themselves seem brave and strong.

Choice B is incorrect because making edelweiss seem more dangerous to reach than other mountain flowers are would likely discourage flower enthusiasts from trying to reach edelweiss rather than encourage them to explore the Swiss Alps. The text suggests that climbers’ self-interested motivation for exaggerating the difficulty of climbing to areas where edelweiss grew was to enhance their own reputations for being especially brave, not to promote tourism in the Swiss Alps. *Choice C* is incorrect because the text states that according to Scheidegger, mountain climbers exaggerated the difficulty of reaching edelweiss for self-interested reasons, not because they wanted to report genuine scientific observations about the flower’s unique characteristics. The text provides no indication that the climbers sincerely believed that their claims about the especially challenging terrain in which edelweiss grew were accurate, much less that they wanted to share this information with scientists. *Choice D* is incorrect because the text doesn’t suggest that mountain climbers were trying to prove that edelweiss were more common in Switzerland than elsewhere. The text’s focus is on the supposed difficulty and danger of reaching the flowers, not on their relative abundance in the Swiss Alps compared to other regions. In fact, the text implies that mountain climbers exaggerated how difficult edelweiss were to reach; it therefore wouldn’t make sense to say that they exaggerated this difficulty to demonstrate that edelweiss were relatively common.

QUESTION 18

Choice A is the best answer because it most logically completes the text’s discussion of modeling the atmosphere of Titan. The text establishes that the methane mole fraction (the ratio of methane to other atmospheric components) influences precipitation, humidity, and other meteorological phenomena and that observational data for Titan are too lacking (“too sparse and discrepant”) to fully determine the range of this measure across the moon’s various atmospheric levels. The text indicates that because of this limitation, researchers developed an atmospheric model (IPSL) that uses a simplified approach: it applies a uniform methane mole fraction for the lowest atmospheric level. Since it is very unlikely that the actual methane distribution across atmospheric levels on Titan is uniform, and since the methane mole fraction directly affects precipitation and humidity, it logically follows that the model’s simplified treatment of the methane mole fraction would lead to some disagreements between the model’s simulations and Titan’s actual precipitation and humidity.

Choice B is incorrect because the text makes it clear that the model *already* uses a single value for Titan’s methane mole fraction instead of a range. Further, the text suggests that the single value is a simplification of what is likely a more complex reality, so there’s no reason to assume that a single value would be more

desirable than a range. *Choice C* is incorrect. The text strongly suggests that Titan's actual methane mole fraction differs from the uniform value used in the model, even though some of the model's outputs align closely with observations, but nothing in the text indicates how the actual value would differ. There's no reason to assume that the actual methane mole fraction is higher than what the model uses. The "lowest level" mentioned in the text refers to the atmosphere and not Titan's methane mole fraction. *Choice D* is incorrect because the text makes it clear that the model *doesn't* reflect the actual variations in the methane mole fraction across Titan's atmospheric levels, since observational data are too limited. Because the model uniformly applies a set methane mole fraction, no inconsistencies in the simulations would be attributable to actual variations in the moon's methane mole fraction.

QUESTION 19

Choice B is the best answer. The convention being tested is end-of-sentence punctuation. This choice correctly uses a period to punctuate a declarative sentence ending with a relative clause that presents an indirect question ("why ganga has been used as a communication method.")

Choice A is incorrect. It's unconventional to use a question mark to punctuate a declarative sentence that presents an indirect question. *Choice C* is incorrect. The structure requires a declarative clause at the end of the sentence, not an interrogative clause that asks a direct question, such as "has ganga been used as a communication method." *Choice D* is incorrect. The structure requires a declarative clause at the end of the sentence, not an interrogative clause that asks a direct question, such as "has ganga been used as a communication method."

QUESTION 20

Choice B is the best answer. The convention being tested is the use of verb forms within a sentence. The nonfinite to-infinitive verb "to create" is correctly used to form a subordinate clause that expresses how ArcGIS facilitates the creation of maps (by analyzing and arranging certain data).

Choice A is incorrect because it results in an ungrammatical sentence. The finite verb "create" can't be used in this way to form a subordinate clause that expresses how ArcGIS facilitates the creation of maps. *Choice C* is incorrect because it results in an ungrammatical sentence. The nonfinite participle "creating" can't be used in this way to form a subordinate clause that expresses how ArcGIS facilitates the creation of maps. *Choice D* is incorrect because it results in an ungrammatical sentence. The finite verb "created" can't be used in this way to form a subordinate clause that expresses how ArcGIS facilitates the creation of maps.

QUESTION 21

Choice B is the best answer. The convention being tested is punctuation use between a verb and its complement. No punctuation is needed between the verb

“is called” and its complement “the chromophore.” The complement helps complete the idea of the verb—in this case, it explains what the part of a compound that determines the compound’s color is called—and any punctuation between the two results in an ungrammatical sentence.

Choice A is incorrect because no punctuation is needed between the verb and its complement. *Choice C* is incorrect because no punctuation is needed between the verb and its complement. *Choice D* is incorrect because no punctuation is needed between the verb and its complement.

QUESTION 22

Choice B is the best answer. The convention being tested is the use of punctuation within a sentence. This choice correctly uses a dash after “cooperation” to pair with the dash after “that” to separate the supplementary element “assuming solar panel installers’ cooperation” from the rest of the sentence.

Choice A is incorrect because a comma can’t be paired with a dash in this way to separate the supplementary element from the rest of the sentence. *Choice C* is incorrect because a colon can’t be paired with a dash in this way to separate the supplementary element from the rest of the sentence. *Choice D* is incorrect because it fails to use appropriate punctuation to separate the supplemental element from the rest of the sentence.

QUESTION 23

Choice B is the best answer. The convention being tested is the use of punctuation within a sentence. The comma after “newts” pairs with the comma after “instance” to separate the supplementary element “for instance” from the rest of the sentence. The supplementary element appears between the prepositional phrase (“In...newts”) and the main clause of the sentence (“you... *orientalis*”) and signals that the sentence is presenting an example of how keys and values function in JSON files.

Choice A is incorrect because it results in a sentence fragment (“In a JSON file storing data about fire belly newts”). *Choice C* is incorrect because a colon can’t be paired with a comma in this way to separate the supplementary element “for instance” from the rest of the sentence. *Choice D* is incorrect because a semicolon can’t be paired with a comma in this way to separate the supplementary element “for instance” from the rest of the sentence.

QUESTION 24

Choice B is the best answer. The convention being tested is subject-modifier placement. This choice makes the noun phrase “amorphous ice” the subject of the sentence and places it immediately after the modifying phrase “defined... structure.” In doing so, this choice clearly establishes that amorphous ice—and not another noun in the sentence—is being described as having a disorganized molecular structure.

Choice A is incorrect because it results in a dangling modifier. The placement of the noun “ice Ih” immediately after the modifying phrase illogically suggests that ice Ih has a disorganized molecular structure, whereas ice Ih is previously described as having molecules that form a hexagonal pattern. *Choice C* is incorrect because it results in a dangling modifier. The placement of the noun phrase “the lack of thermal energy” immediately after the modifying phrase illogically suggests that the lack of thermal energy has a disorganized molecular structure. *Choice D* is incorrect because it results in a dangling modifier. The placement of “ice Ih” immediately after the modifying phrase illogically suggests that ice Ih has a disorganized molecular structure, whereas ice Ih is previously described as having molecules that form a hexagonal pattern.

QUESTION 25

Choice A is the best answer. The convention being tested is subject-modifier placement. This choice makes the noun “electrograms” the subject of the sentence and places it immediately after the modifying phrase “recordings of electrical activity in the brain.” In doing so, this choice clearly establishes that electrograms—and not other nouns or noun phrases in the sentence—are recordings of electrical activity in the brain.

Choice B is incorrect because it results in a dangling modifier. The placement of the noun phrase “the most highly skilled soccer players” immediately after the modifying phrase illogically suggests that the players are recordings of electrical activity in the brain. *Choice C* is incorrect because it results in a dangling modifier. The placement of the noun phrase “responses to hypothetical match scenarios” immediately after the modifying phrase illogically suggests that these responses are recordings of electrical activity in the brain. *Choice D* is incorrect because it results in a dangling modifier. The placement of the noun phrase “hypothetical match scenario responses” immediately after the opening phrase illogically suggests that these responses are recordings of electrical activity in the brain.

QUESTION 26

Choice B is the best answer. The convention being tested is the use of verb forms within a sentence. The nonfinite present participle “searching” is correctly used to form a supplementary element that modifies the subject “scientists,” indicating that the scientists who adapted the test were looking for evidence of self-awareness in snakes.

Choice A is incorrect because it results in an ungrammatical sentence. The finite past tense verb “searched” can’t be used in this way to form a supplementary element that indicates what evidence the scientists who adapted the test were looking for. *Choice C* is incorrect because it results in an ungrammatical sentence. The finite past progressive verb “were searching” can’t be used in this way to form a supplementary element that indicates what evidence the scientists who adapted the test were looking for. *Choice D* is incorrect because it results in an ungrammatical sentence. The finite present perfect verb “have searched” can’t be used in this way to form a supplementary element that indicates what evidence the scientists who adapted the test were looking for.

QUESTION 27

Choice D is the best answer. “Finally” logically signals that the information in this sentence—that the artisan slips paper in and out of the liquid to transfer the design onto paper—is the final step in the process of Turkish paper marbling, which began with what happens “first” (filling a tray with a solution) and continued with what happens “next” (adding inks or paints to the solution).

Choice A is incorrect because “actually” illogically signals that the information in this sentence about slipping paper in and out of liquid to transfer a design onto paper is unexpected in light of the previously described steps. Instead, the sentence describes the final step in the process of Turkish paper marbling.

Choice B is incorrect because “therefore” illogically signals that the information in this sentence about slipping paper in and out of liquid to transfer a design onto paper is a consequence of the previously described steps. Instead, the sentence describes the final step in the process of Turkish paper marbling. *Choice C* is incorrect because “nevertheless” illogically signals that the information in this sentence about slipping paper in and out of liquid to transfer a design onto paper contrasts with the previously described steps. Instead, the sentence describes the final step in the process of Turkish paper marbling.

QUESTION 28

Choice B is the best answer. “Instead” logically signals that the information about the researchers’ discovery in this sentence—that jorō spiders are gentle giants who “freeze” in place when disturbed—contradicts their initial hypothesis about the spiders’ aggressiveness described in the previous sentence.

Choice A is incorrect because “therefore” illogically signals that the discovery about jorō spiders’ gentleness is a result of the initial hypothesis about their aggressive behavior. Instead, the sentence presents information that contradicts the initial hypothesis. *Choice C* is incorrect because “for example” illogically signals that the discovery about jorō spiders’ gentleness exemplifies the initial hypothesis about their aggressive behavior. Instead, the sentence presents information that contradicts the initial hypothesis. *Choice D* is incorrect because “in other words” illogically signals that the discovery about jorō spiders’ gentleness is merely restating or rephrasing the initial hypothesis about their aggressive behavior. Instead, the sentence presents information that contradicts the initial hypothesis.

QUESTION 29

Choice C is the best answer. “Indeed” logically signals that the information that follows—that the majority of jets reach heights of only 20 to 50 km—offers additional emphasis in support of the previous claim that jets reaching the ionosphere (about 80 km above Earth) are outliers.

Choice A is incorrect because “nevertheless” illogically signals that the information about most jets reaching heights of only 20 to 50 km contrasts with the previous claim that jets reaching the ionosphere are outliers. Instead, it provides additional emphasis in support of that claim. *Choice B* is incorrect because “consequently” illogically signals that the information about most jets

reaching heights of only 20 to 50 km is a consequence, or result, of some jets being outliers. Instead, it offers additional emphasis in support of the claim that jets reaching the ionosphere are outliers. *Choice D* is incorrect because “in addition” illogically signals that the information about most jets reaching heights of only 20 to 50 km merely adds to the previous claim that jets reaching the ionosphere are considered outliers. Instead, it provides additional emphasis in support of that claim.

QUESTION 30

Choice B is the best answer. “Ultimately” logically signals that the information in this sentence—that, for Dillard, nature’s mesmerizing intricacy and pitiless harshness are inextricably linked—is the final conclusion or realization reached after her struggle to reconcile the juxtapositions of the natural world mentioned in the previous sentence.

Choice A is incorrect because “to that end” illogically signals that linking nature’s intricacy and harshness was Dillard’s deliberate goal or purpose in struggling to reconcile nature’s juxtapositions. Instead, the sentence presents her final realization after that struggle. *Choice C* is incorrect because “moreover” illogically signals that the information in this sentence merely adds to Dillard’s struggle to reconcile nature’s juxtapositions mentioned in the previous sentence. Instead, the sentence presents her final realization after that struggle. *Choice D* is incorrect because “hence” illogically signals that nature’s intricacy and harshness being linked is a direct consequence of Dillard’s struggle to reconcile the juxtapositions mentioned in the previous sentence. Instead, the sentence presents her final realization after that struggle.

QUESTION 31

Choice C is the best answer. The sentence presents the methods used in the study, noting that researchers tagged 138 female wood ducks with radio frequency ID trackers and recorded the number of nest boxes each duck visited.

Choice A is incorrect. While the sentence mentions an aspect of the study’s design (that the researchers recorded the number of nest boxes the wood ducks visited), it primarily focuses on a finding of the study rather than the methods the researchers used. *Choice B* is incorrect. While the sentence mentions an aspect of the study’s design (that the researchers tracked the number of nest boxes the wood ducks visited), it primarily focuses on a finding of the study rather than the methods the researchers used. *Choice D* is incorrect. The sentence misrepresents information from the notes. The researchers used radio frequency ID trackers to record the ducks’ visits; they didn’t investigate each site to look for evidence that it had been visited.

QUESTION 32

Choice A is the best answer. The sentence emphasizes a similarity between the ages of the two pedestrian malls, noting that both are relatively old—Qianmen Dajie has roots that go back hundreds of years and Rue Mouffetard has existed for centuries.

Choice B is incorrect. The sentence emphasizes that both locations are famous pedestrian malls and notes that they are in different locations; it doesn't emphasize a similarity in their ages. *Choice C* is incorrect. While the sentence emphasizes that both locations are pedestrian malls, it doesn't emphasize a similarity in their ages. *Choice D* is incorrect. The sentence explains that Qianmen Dajie and Rue Mouffetard are examples of pedestrian malls and gives information about pedestrian malls in general; it doesn't emphasize a similarity in the ages of these two malls specifically.

QUESTION 33

Choice A is the best answer. The sentence compares Fodor's hypothesis with Carruthers's, noting that because Fodorian modularity considers some but not all cognitive systems modular, it is not as expansive in its definition of modularity as Carruthers's MMH, which includes all cognitive systems.

Choice B is incorrect. The sentence describes Carruthers's hypothesis—that modularity of mind includes all cognitive systems—but indicates only that this hypothesis followed Fodor's 1983 hypothesis rather than making a comparison between the hypotheses. *Choice C* is incorrect. The sentence misrepresents a difference between Fodor's and Carruthers's hypotheses. According to the information in the notes, both Fodor and Carruthers consider low-level cognitive systems modular, and their difference lies in whether they also consider high-level systems modular. *Choice D* is incorrect. The sentence misrepresents a difference between Fodor's and Carruthers's hypotheses. According to the information in the notes, both Fodor and Carruthers consider the mind to be at least partly composed of innate neural structures (modules). Their difference lies in whether they consider high-level systems modular.

Math

Module 1

(27 questions)

QUESTION 1

Choice D is correct. The sum of the angle measures of a triangle is 180° . It's given that the measure of angle Q is 132° , the measure of angle P is 24° , and the measure of angle R is x° . It follows that $x + 132 + 24 = 180$, or $x + 152 = 180$. Subtracting 152 from both sides of this equation yields $x = 24$.

Alternate approach: It's given that in the triangle shown, $PQ = QR$. Thus, triangle PQR is an isosceles triangle. Since base angles of an isosceles triangle are congruent, the measure of angle P is equal to the measure of angle R . Therefore, the measure of angle R is 24° , and the value of x is 24.

Choice A is incorrect. This is the sum of the measures, in degrees, of angles Q and P , not the measure, in degrees, of angle R . **Choice B** is incorrect and may result from conceptual or calculation errors. **Choice C** is incorrect. This is the sum of the measures, in degrees, of angles P and R , not the measure, in degrees, of angle R .

QUESTION 2

Choice A is correct. Subtracting 1 from both sides of the given equation, $4x + 1 = 33$, yields $4x = 32$. Therefore, $4x = 32$ has the same solution as the given equation.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 3

Choice C is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m is the slope of the graph of $y = f(x)$ in the xy -plane and $(0, b)$ is the y -intercept of the graph. It's given that for the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7. Therefore, $m = 7$.

It's also given that the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 5)$. Therefore, the y -intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting 7 for m and 5 for b in the equation $f(x) = mx + b$ yields $f(x) = 7x + 5$.

Choice A is incorrect. This equation defines a function whose graph has a slope of 5, not 7, and passes through the point $(0, 0)$, not $(0, 5)$. *Choice B* is incorrect. This equation defines a function whose graph has a slope of 35, not 7, and passes through the point $(0, 0)$, not $(0, 5)$. *Choice D* is incorrect. This equation defines a function whose graph has a slope of 12, not 7.

QUESTION 4

Choice A is correct. Adding 40 to both sides of the given equation yields $8x^2 = 72$. Dividing both sides of this equation by 8 yields $x^2 = 9$. Taking the square root of both sides of this equation yields $x = 3$ or $x = -3$. Therefore, the positive solution to the given equation is 3.

Choice B is incorrect. This is the solution to the equation $8x = 32$, not the given equation. *Choice C* is incorrect. This is the solution to the equation $8x - 40 = 32$, not the given equation. *Choice D* is incorrect. This is the solution to the equation $x - 40 = 32$, not the given equation.

QUESTION 5

Choice B is correct. If one of these children is selected at random, the probability of selecting a child who chose a vegetarian sandwich is equal to the number of children who chose a vegetarian sandwich divided by the total number of children. According to the table, there are a total of 50 children, and 9 of these children chose a vegetarian sandwich. Therefore, the probability of selecting a child who chose a vegetarian sandwich is $\frac{9}{50}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 6

The correct answer is 75. It's given that Amara harvested 750 cherry tomatoes and gave 10% of them to her neighbor. 10% can be written as $\frac{10}{100}$, or 0.1.

Therefore, the number of cherry tomatoes that Amara gave to her neighbor can be calculated by multiplying the total number of cherry tomatoes by 0.1, which gives $(0.1)(750)$, or 75.

QUESTION 7

The correct answer is 30. It's given that $x + y = 125$. Substituting 125 for $x + y$ in the second equation of the given system yields $125 + y = 155$. Subtracting 125 from both sides of this equation yields $y = 30$. Therefore, the value of y is 30.

QUESTION 8

Choice A is correct. It's given that each participant earns 1 point for each game the participant plays that ends in a draw and 3 points for each game the participant wins. Since d represents the number of games that a certain participant has played that ended in a draw and w represents the number of games this participant has won, this participant has earned a total of $1d + 3w$, or $d + 3w$, points. Since this participant earned 41 points, the equation $d + 3w = 41$ represents this situation.

Choice B is incorrect. This equation represents the situation where the participant earned 3 points for each game played that ended in a draw and 1 point for each game won and earned 41 points. **Choice C** is incorrect. This equation represents the situation where the participant earned 1 point for each game played that ended in a draw and $\frac{1}{3}$ point for each game won and earned 41 points. **Choice D** is incorrect. This equation represents the situation where the participant earned $\frac{1}{3}$ point for each game played that ended in a draw and 1 point for each game won and earned 41 points.

QUESTION 9

Choice C is correct. It's given that function g is defined by $g(x) = \sqrt{x} + 300$. Substituting 81 for x in function g yields $g(81) = \sqrt{81} + 300$, which is equivalent to $g(81) = 9 + 300$, or $g(81) = 309$. Therefore, the value of $g(x)$ when $x = 81$ is 309.

Choice A is incorrect. This is the value of $\sqrt{81}$, not $\sqrt{81} + 300$. **Choice B** is incorrect and may result from conceptual or calculation errors. **Choice D** is incorrect. This is the value of $81 + 300$, not $\sqrt{81} + 300$.

QUESTION 10

Choice C is correct. It's given that 8 out of 300 customers in a random sample said they would be interested in the new plan. This means the fraction of sampled customers who said they would be interested is $\frac{8}{300}$. Multiplying this fraction by the total customer population yields $(\frac{8}{300})(30,000)$, or 800. Therefore, the best estimate of the total number of customers who would be interested in the new service plan is 800.

Choice A is incorrect. This is the number of customers surveyed who said they would be interested in the new plan, not the best estimate of the total number of customers who would be interested in the new plan. **Choice B** is incorrect and may result from conceptual or calculation errors. **Choice D** is incorrect and may result from conceptual or calculation errors.

QUESTION 11

Choice D is correct. For the expression $64t^2s^3 - 56t^3s$, 4 is a common factor of 64 and 56, t^2 is a common factor of t^2 and t^3 , and s is a common factor of s^3 and s . It follows that $4t^2s$ is a common factor of each term in the given

expression. Factoring out $4t^2s$ from each term in the expression $64t^2s^3 - 56t^3s$ yields $4t^2s(16s^2 - 14t)$. Therefore, $4t^2s(16s^2 - 14t)$ is equivalent to $64t^2s^3 - 56t^3s$.

Choice A is incorrect. This expression is equivalent to $64ts^3 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$. *Choice B* is incorrect. This expression is equivalent to $64t^3s^3 - 56t^2s$, not $64t^2s^3 - 56t^3s$. *Choice C* is incorrect. This expression is equivalent to $64t^3s^2 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$.

QUESTION 12

Choice B is correct. The point (x, y) at which the graphs of the given equations intersect is the solution to the system of equations. Subtracting 5 from both sides of the equation $x + 5 = 14$ yields $x = 9$. Substituting 9 for x in the equation $y = 4x^2 + 4$ yields $y = 4(9)^2 + 4$, or $y = 328$. Therefore, the graphs of the equations in the given system intersect at the point $(9, 328)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 13

The correct answer is 13. It's given that the equation $8x + 11y = 170$ gives the possible combinations of the number of 2009 premium grade Log Cabin Pennies, x , and the number of 1996 select grade Lincoln Pennies, y , in a collection that is worth a total of \$170. It's also given that there are 6 1996 select grade Lincoln Pennies in the collection, and thus $y = 6$. Substituting 6 for y in the given equation yields $8x + 11(6) = 170$, or $8x + 66 = 170$. Subtracting 66 from both sides of this equation yields $8x = 104$. Dividing both sides of this equation by 8 yields $x = 13$. Therefore, there are 13 2009 premium grade Log Cabin Pennies in the collection.

QUESTION 14

The correct answer is 15,000. Let y represent the population of Smithville x years after 1659. Since the population doubled every 75 years, the relationship between x and y can be represented by an exponential equation in the form

$y = a(b)^{\frac{x}{k}}$, where a is the population of Smithville in 1659 and the population increases by a factor of b every k years. It's given that the population doubled, or increased by a factor of 2, every 75 years. Therefore, $b = 2$ and $k = 75$.

Substituting 2 for b and 75 for k in the equation $y = a(b)^{\frac{x}{k}}$ yields $y = a(2)^{\frac{x}{75}}$. It's also given that the population of this town was 240,000 in 1959. The number of years from 1659 to 1959 is $1959 - 1659$, or 300 years. Therefore, when $x = 300$, the value of y is 240,000. Substituting 300 for x and 240,000 for y in the equation $y = a(2)^{\frac{x}{75}}$ yields $240,000 = a(2)^{\frac{300}{75}}$, or $240,000 = a(2)^4$, which is equivalent to $240,000 = a(16)$. Dividing both sides of this equation by 16 yields $15,000 = a$. Therefore, the population of this town in 1659 was 15,000.

QUESTION 15

Choice A is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 3.2)$ and $(2, 4.4)$. It follows that the slope of the line of best fit is approximately $\frac{4.4 - 3.2}{2 - 0}$, which is equivalent to $\frac{1.2}{2}$, or 0.6. Therefore, of the given choices, 0.6 is closest to the slope of the line of best fit shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 16

Choice A is correct. Taking the square root of both sides of $(x + 15)^2 = 0$ yields $x + 15 = 0$. Subtracting 15 from both sides of this equation yields $x = -15$. Therefore, this equation has exactly one distinct real solution.

Choice B is incorrect. This equation has no distinct real solutions, not exactly one distinct real solution. *Choice C* is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution. *Choice D* is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

QUESTION 17

Choice C is correct. It's given that the cost to rent a bus is \$950 for the first 3 hours and an additional \$50 per hour for each hour after the first 3 hours. It's also given that t represents the total number of hours and $t > 3$. Therefore, the number of additional hours after the first 3 hours can be represented by the expression $t - 3$. The cost for these additional hours is \$50 per hour, so the cost for the additional hours can be represented by the expression $50(t - 3)$. The total cost can be calculated by adding the cost for the first 3 hours to the cost for the additional hours and can be represented by the expression $950 + 50(t - 3)$. It's also given that the total cost to rent the bus from the company for t hours is \$1,150. Thus, the equation that represents this situation is $950 + 50(t - 3) = 1,150$.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 18

Choice B is correct. The intercepts of the graph of each equation can be found by substituting 0 for one variable at a time. For the equation $x + y = 53$, substituting 0 for y yields $x = 53$, so the x -intercept of the graph of the equation is $(53, 0)$. Substituting 0 for x in this equation yields $y = 53$, so the y -intercept of the graph of the equation is $(0, 53)$. Thus, the graph of the equation $x + y = 53$ has intercepts at $(53, 0)$ and $(0, 53)$. For the equation $11x + 18y = 730$, substituting 0

for y yields $x = \frac{730}{11}$, so the x -intercept of the graph of the equation is approximately $(66, 0)$. Substituting 0 for x in this equation yields $y = \frac{730}{18}$, so the y -intercept of the graph of the equation is approximately $(0, 41)$. Thus, the graph of the equation $11x + 18y = 730$ has intercepts at approximately $(66, 0)$ and $(0, 41)$. Choice B is a graph that shows one line with intercepts at $(0, 53)$ and $(53, 0)$ and another line with intercepts at approximately $(0, 41)$ and $(66, 0)$. Thus, choice B represents this situation.

Choice A is incorrect. This graph represents the equations $y = x$ and $11x + 18y = 730$. Choice C is incorrect. This graph represents the equations $x + y = 53$ and $-11x + 18y = 730$. Choice D is incorrect. This graph represents the equations $y = 53$ and $-11x + 18y = 730$.

QUESTION 19

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 18. Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS , an expression representing QS can be found by substituting 18 for RS in this equation, which yields $\sin Q = \frac{18}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 18$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{18}{\sin Q}$. Therefore, the expression $\frac{18}{\sin Q}$ represents the length of QS .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

QUESTION 20

The correct answer is 100. An equation of a circle in the xy -plane can be written as $(x - t)^2 + (y - u)^2 = r^2$, where the center of the circle is (t, u) , the radius of the circle is r , and where t , u , and r are constants. It's given that the equation of circle A is $(x + 5)^2 + (y - 5)^2 = 25$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 5^2$. Therefore, the center of circle A is $(-5, 5)$ and the radius of circle A is 5. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $(-5, 5)$ and the radius of circle B is $2(5)$, or 10. Substituting -5 for t , 5 for u , and 10 for r in the equation $(x - t)^2 + (y - u)^2 = r^2$ yields $(x + 5)^2 + (y - 5)^2 = 10^2$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 100$. It follows that the equation of circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = 100$. It's also given that the equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. Therefore, the value of k is 100.

QUESTION 21

The correct answer is 29. The solutions to a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be calculated using the quadratic formula, which gives two solutions: $x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 1$, $b = 7$, and $c = 5$. It follows that one solution to the given quadratic equation is $x = \frac{-7 + \sqrt{7^2 - 4(1)(5)}}{2(1)}$, or $x = \frac{-7 + \sqrt{49 - 20}}{2}$, which is equivalent to $x = \frac{-7 + \sqrt{29}}{2}$. It's given that one solution to the given equation can be written as $x = \frac{-7 + \sqrt{k}}{2}$, where k is a constant. It follows that the value of k is 29.

QUESTION 22

Choice B is correct. The mean of a data set is calculated by dividing the sum of the values in the data set by the number of values in the data set. It's given that the mean length of the 240 gray seals from Muskeget Island was 88 inches. This can be represented by the equation $\frac{x}{240} = 88$, where x represents the sum of the lengths, in inches, of the 240 gray seals from Muskeget Island. Multiplying both sides of this equation by 240 yields $x = (240)(88)$, or $x = 21,120$ inches. It's also given that the mean length of the 120 gray seals from Sable Island was 94 inches. This can be represented by the equation $\frac{y}{120} = 94$, where y represents the sum of the lengths, in inches, of the 120 gray seals from Sable Island. Multiplying both sides of this equation by 120 yields $y = (120)(94)$, or $y = 11,280$ inches. Therefore, the sum of the lengths, in inches, of all 360 gray seals is $21,120 + 11,280$, or 32,400. Dividing this sum by 360 yields $\frac{32,400}{360}$, or 90 inches. Therefore, the mean length of all 360 gray seals the scientist measured for this study is 90 inches.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the mean length of 88 inches and 94 inches, not the mean length of all 360 gray seals the scientist measured for this study. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 23

Choice B is correct. The graph of $y = f(x) + 4$ is shown, and the graph passes through the point $(0, 8)$. Substituting 0 for x and 8 for y in this equation yields $8 = f(0) + 4$. Subtracting 4 from both sides of this equation yields $4 = f(0)$. Therefore, when $x = 0$, the value of $f(x)$ is 4. Each of the given choices is in the form $f(x) = -3^x + k$, where k is a constant. Substituting 0 for x and 4 for $f(x)$ in this equation yields $4 = -3^0 + k$, or $4 = -1 + k$. Adding 1 to both sides of this equation yields $5 = k$. Substituting 5 for k in the equation $f(x) = -3^x + k$ yields $f(x) = -3^x + 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 24

Choice A is correct. It's given that two lines intersect at exactly one point, forming two acute angles and two obtuse angles. When two lines intersect, opposite angles are congruent and adjacent angles are supplementary. It's given that the measure of one of the angles is $(9x - 140)^\circ$, so the measure of the opposite angle is $(9x - 140)^\circ$, and the measure of each adjacent angle is $(180 - (9x - 140))^\circ$, or $(-9x + 320)^\circ$. The possible sums of the measures of any two of the angles are $(9x - 140)^\circ + (9x - 140)^\circ$, or $(18x - 280)^\circ$, $(-9x + 320)^\circ + (-9x + 320)^\circ$, or $(-18x + 640)^\circ$, and $(9x - 140)^\circ + (-9x + 320)^\circ$, or 180° . Of the given choices, only $(-18x + 280)^\circ$ is not equivalent to any of these expressions and could not be the sum of the measures of any two of the angles.

Choice B is incorrect. This is the sum of the measures of two angles that are adjacent to the given angle. **Choice C** is incorrect. This is the sum of the measure of the given angle and the measure of its opposite angle. **Choice D** is incorrect. This is the sum of the measure of the given angle and the measure of an angle adjacent to the given angle.

QUESTION 25

Choice C is correct. A system of two linear equations in two variables has at least one solution if the equations represent the same line or lines that intersect at exactly one point. Statement I gives the equation $3x + 13.5y = 10.5$. Multiplying both sides of the given equation, $2x + 9y = 7$, by $\frac{3}{2}$ yields $3x + 13.5y = 10.5$, so the equation in statement I is equivalent to the given equation. Therefore, the system formed by these two equations represents the same line and has infinitely many solutions. Statement II gives the equation $3x - 13.5y = 10.5$. Adding the right- and left-hand sides of $3x + 13.5y = 10.5$ and $3x - 13.5y = 10.5$ yields $3x + 13.5y + 3x - 13.5y = 10.5 + 10.5$, or $6x = 21$. Dividing both sides of this equation by 6 yields $x = \frac{21}{6}$. Therefore, the lines represented by these equations intersect at exactly one point, where $x = \frac{21}{6}$, and the system formed by the given equation and the equation in statement II has one solution. It follows that both the equation in statement I and the equation in statement II could be the other equation in a system of equations with at least one solution.

Choice A is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution. **Choice B** is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution. **Choice D** is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

QUESTION 26

Choice B is correct. It's given that the base area of the right rectangular prism is $24t \text{ cm}^2$ and the length of the base is $\frac{8}{3} \text{ cm}$. Dividing the base area by the length yields the width: $\frac{24t}{\frac{8}{3}}$, or $(24t)(\frac{3}{8})$, or $9t \text{ cm}$. It's also given that the height of the prism is 15 cm . A right rectangular prism has two rectangular bases and four rectangular lateral faces. The total area of the two bases is $2(24t)$, or $48t \text{ cm}^2$. The four lateral faces include two with dimensions $\frac{8}{3} \text{ cm}$ by 15 cm and two with dimensions $9t \text{ cm}$ by 15 cm . The total area of these four lateral faces is $2(\frac{8}{3})(15) + 2(9t)(15) \text{ cm}^2$, which is equivalent to $80 + 270t \text{ cm}^2$. Adding this to the total area of the two bases yields $48t + 270t + 80$, or $318t + 80 \text{ cm}^2$. Thus, the expression, which represents the surface area, in cm^2 , of the right rectangular prism is $318t + 80$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the volume, in cm^3 , not the surface area, in cm^2 .

QUESTION 27

The correct answer is 3,331. For engine speeds between 1,000 revolutions per minute (rpm) and 6,000 rpm, it's given that the equation $f(x) = \frac{1}{7}(x - a) + 433$ defines the linear function f and gives the predicted power, in brake horsepower (bhp), at an engine speed of x rpm, where a is a constant. It's also given that the car's predicted power is 433 bhp at an engine speed of 3,331 rpm. Substituting 3,331 for x and 433 for $f(x)$ in the equation yields $433 = \frac{1}{7}(3,331 - a) + 433$. Subtracting 433 from both sides of this equation yields $0 = \frac{1}{7}(3,331 - a)$. Multiplying both sides of this equation by 7 yields $0 = 3,331 - a$. Adding a to both sides of this equation yields $a = 3,331$. Thus, the value of a is 3,331.

Math

Module 2

(27 questions)

QUESTION 1

Choice C is correct. In the given expression, $6x$ and $5x$ are like terms. Combining these like terms yields $11x$. It follows that the expression $6x + 5x + 4y$ is equivalent to $11x + 4y$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 2

Choice B is correct. The dot plot gives the diameter, to the nearest inch, of each sea star in a group of tide pools. The number of dots above a given diameter represents the number of sea stars with that diameter, to the nearest inch. There are 6 dots above 16 inches. Therefore, 6 sea stars had a diameter, to the nearest inch, of 16 inches.

Choice A is incorrect and may result from calculation or conceptual errors.

Choice C is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of 17 inches, not 16 inches. *Choice D* is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of 18 inches, not 16 inches.

QUESTION 3

Choice D is correct. The area, A , of a rectangle can be found using the formula $A = \ell w$, where ℓ represents the length and w represents the width of the rectangle. It's given that a rectangle has a length of 56 inches and a width of 28 inches. Substituting 56 for ℓ and 28 for w in the formula $A = \ell w$ yields $A = (56)(28)$, or $A = 1,568$. Therefore, the area, in square inches, of the rectangle is 1,568.

Choice A is incorrect. This is the width, in inches, not the area, in square inches, of the rectangle. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 4

Choice D is correct. Dividing both sides of the equation $10x = 110$ by 10 yields $x = 11$. Substituting 11 for x in the equation $6x - 63 = y$ yields $6(11) - 63 = y$, which is equivalent to $66 - 63 = y$, or $3 = y$. Therefore, the value of y is 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not y . *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 5

Choice A is correct. Substituting 63 for $f(x)$ in the given function yields $63 = 9(2x + 3)$. Dividing both sides of this equation by 9 yields $7 = 2x + 3$. Subtracting 3 from both sides of this equation yields $4 = 2x$. Dividing both sides of this equation by 2 yields $2 = x$. Therefore, $f(x) = 63$ when the value of x is 2.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 6

The correct answer is 8.6. Dividing both sides of the given equation by 10 yields $x = 8.6$. Therefore, 8.6 is the solution to the given equation. Note that 8.6 and $43/5$ are examples of ways to enter a correct answer.

QUESTION 7

The correct answer is 3,600. The predicted number of bacteria initially measured in the growth medium is the value of y when $x = 0$. Substituting 0 for x in the given equation yields $y = 3,600(a)^0$. Since a is positive, this equation is equivalent to $y = 3,600(1)$. Therefore, when $x = 0$, the value of y is 3,600. Thus, the predicted number of bacteria initially measured in the growth medium is 3,600.

QUESTION 8

Choice A is correct. It's given that each container costs \$1.87 and c represents the number of containers. It follows that $1.87c$ represents the cost, in dollars, of c containers. It's also given that each roll of tape costs \$2.40 and t represents the number of rolls of tape. It follows that $2.40t$ represents the cost, in dollars, of t rolls of tape. Therefore, the total cost, in dollars, for c containers and t rolls of tape is $1.87c + 2.40t$. It's given that Leo has \$15. It follows that the total cost, in dollars, must be less than or equal to 15. Therefore, the inequality $1.87c + 2.40t \leq 15$ represents this relationship.

Choice B is incorrect. This inequality represents a relationship where Leo has at least \$15, not at most \$15. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 9

Choice B is correct. The graph of function f shows that as x increases, $f(x)$ also increases, which means that f is an increasing function. The graph of function f is curved upward and increases at an increasing rate, not a constant rate. This behavior is characteristic of exponential functions, not linear functions. Therefore, of the given choices, function f is best described as increasing exponential.

Choice A is incorrect. For a decreasing function, as x increases, $f(x)$ decreases rather than increases. *Choice C* is incorrect. For a decreasing function, as x increases, $f(x)$ decreases rather than increases, and the graph of a linear function isn't curved. *Choice D* is incorrect. The graph of a linear function isn't curved.

QUESTION 10

Choice D is correct. The equation for the line representing the boundary of the shaded region can be written in the form $y = b$, where $(0, b)$ is the y -intercept of the line in the xy -plane. For the graph shown in the xy -plane, the boundary line passes through the points $(-15, 36)$ and $(0, 36)$. Thus, the equation of the boundary line is $y = 36$. Since the shaded region represents all the points on and above this boundary line, it follows that the graph represents all the solutions to the inequality $y \geq 36$.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors.

QUESTION 11

Choice C is correct. It's given that one of the two players in a match is eliminated during each round. Therefore, the number of players decreases by half at the end of each round, and so the number of players eliminated at the end of a round also decreases by half. A decreasing exponential equation can be written in the form $p = a(1 - b)^r$, where b is the value of the decrease each round, r , and a is the value of p , the number of players, when $r = 0$. It's given that there are 240 players in the tennis competition. Therefore, $a = 240$. Since the number of players eliminated at the end of a round decreases by half, it follows that $b = \frac{1}{2}$. Substituting 240 for a and $\frac{1}{2}$ for b in the equation $p = a(1 - b)^r$ yields $p = 240\left(1 - \frac{1}{2}\right)^r$, or $p = 240\left(\frac{1}{2}\right)^r$.

Choice A is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are 15, not 240, players.

Choice B is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are 15, not 240, players, and for which the number of players doubles, rather than decreases by half, after each

round. *Choice D* is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which the number of players doubles, rather than decreases by half, after each round.

QUESTION 12

Choice A is correct. An equation that defines a line in the xy -plane can be written as $y = mx + b$, where m is the slope of the line and $(0, b)$ is its y -intercept. It's given that line k is defined by the equation $y = 6x + 4$; therefore, the slope of line k is 6. Since line j is parallel to line k in the xy -plane and parallel lines have equal slopes, it follows that the slope of line j is 6. It's also given that line j passes through the point $(0, 5)$, which is its y -intercept. Substituting 6 for m and 5 for b in the equation $y = mx + b$ yields $y = 6x + 5$. Therefore, the equation $y = 6x + 5$ defines line j .

Choice B is incorrect. This equation defines a line that has a slope of -5 , not 6.

Choice C is incorrect. This equation defines a line that has a slope of -6 , not 6.

Choice D is incorrect. This equation defines a line that has a slope of 5, not 6.

QUESTION 13

The correct answer is 45. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of 15. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are 45° . It follows that the value of x is 45.

QUESTION 14

The correct answer is 13. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the radius of the circle. It's given that the circle in the xy -plane is defined by $(x + 2)^2 + (y + 5)^2 = 169$. Therefore, $r^2 = 169$. Taking the positive square root of both sides of this equation yields $r = 13$. Thus, the radius of the circle is 13.

QUESTION 15

Choice B is correct. It's given that the graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole. It follows that for the point $(149.02, 186.05)$, 149.02 represents the molecular weight, in grams per mole, of a normal paraffin and 186.05 represents its estimated boiling point, in degrees Celsius. Therefore, the best interpretation of the point $(149.02, 186.05)$ is that a normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.

Choice A is incorrect. This statement represents a situation where x , not y , represents the boiling point, in degrees Celsius, and y , not x , represents the molecular weight, in grams, per mole. *Choice C* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 16

Choice C is correct. It's given that f is a polynomial function and the graph of $y = f(x)$ in the xy -plane passes through the points $(-5, 0)$, $(1, 0)$, and $(4, 0)$. A point in the form $(a, 0)$ on the graph of $f(x)$ means that $x - a$ is a factor of $f(x)$. Since the graph of the function f passes through the point $(1, 0)$, $x - 1$ must be a factor of $f(x)$.

Choice A is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(-1, 0)$. **Choice B** is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(-4, 0)$. **Choice D** is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(5, 0)$.

QUESTION 17

Choice D is correct. It's given that for the linear function g , the table shows four values of x and their corresponding values of $g(x)$. It's also given that the function can be written as $g(x) = mx + b$, where m and b are constants. The table shows that when the value of x is 1, the corresponding value of $g(x)$ is 32. Substituting 1 for x and 32 for $g(x)$ in $g(x) = mx + b$ yields $32 = m(1) + b$, or $32 = m + b$. Subtracting b from both sides of this equation yields $32 - b = m$. The table also shows that when the value of x is 2, the corresponding value of $g(x)$ is 28. Substituting 2 for x and 28 for $g(x)$ in $g(x) = mx + b$ yields $28 = m(2) + b$, or $28 = 2m + b$. Substituting $32 - b$ for m in this equation yields $28 = 2(32 - b) + b$. Applying the distributive property to the right-hand side of this equation yields $28 = 64 - 2b + b$, or $28 = 64 - b$. Subtracting 64 from both sides of this equation yields $-36 = -b$. Dividing both sides of this equation by -1 yields $36 = b$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

QUESTION 18

Choice B is correct. According to the table, $f(24) = -8$, $f(30) = -8$, and $f(32) = 8$.

Let the quadratic function g be defined by $g(x) = f(x) + 8$. Therefore,

$g(24) = f(24) + 8$, $g(30) = f(30) + 8$, and $g(32) = f(32) + 8$. Substituting -8 for $f(24)$ in the equation $g(24) = f(24) + 8$ yields $g(24) = -8 + 8$, or $g(24) = 0$.

Substituting -8 for $f(30)$ in the equation $g(30) = f(30) + 8$ yields $g(30) = -8 + 8$, or $g(30) = 0$. Substituting 8 for $f(32)$ in the equation $g(32) = f(32) + 8$ yields

$g(32) = 8 + 8$, or $g(32) = 16$. By the factor theorem, if $g(24) = 0$ and $g(30) = 0$,

then $g(x) = k(x - 24)(x - 30)$, where k is a constant. Since $g(32) = 16$,

substituting 32 for x and 16 for $g(x)$ in the equation $g(x) = k(x - 24)(x - 30)$ yields $16 = k(32 - 24)(32 - 30)$, or $16 = k(8)(2)$, which is equivalent to $16 = 16k$.

Dividing both sides of this equation by 16 yields $1 = k$. Substituting 1 for k in the equation $g(x) = k(x - 24)(x - 30)$ yields $g(x) = (x - 24)(x - 30)$. Since

$g(x) = f(x) + 8$, it follows that $g(x) - 8 = f(x)$. Subtracting 8 from both sides of the equation $g(x) = (x - 24)(x - 30)$ yields $g(x) - 8 = (x - 24)(x - 30) - 8$. Substituting $f(x)$ for $g(x) - 8$ in this equation yields $f(x) = (x - 24)(x - 30) - 8$.

Choice A is incorrect. If this equation defined f , then when $x = 24$, the value of $f(x)$ would be 4, not -8 . *Choice C* is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be 32, not 8. *Choice D* is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be -32 , not 8.

QUESTION 19

Choice A is correct. The x -intercept of the graph shown represents the situation in which the cluster contains only M-type stars. Based on the graph, the x -intercept is approximately $(158, 0)$. Therefore, if there are 0 K-type stars in the cluster, then there are approximately 158 M-type stars in the cluster. It's given that the total mass of the stars in this cluster is 127,882 quettagrams. Therefore, the estimated mass of each M-type star is approximately $\frac{127,882}{158}$ quettagrams, or approximately 809.38 quettagrams. Thus, of the given choices, 811 is the closest to the estimated mass, in quettagrams, of each M-type star in this cluster.

Choice B is incorrect. This is the value closest to the estimated mass, in quettagrams, of each K-type star, not each M-type star, in this cluster. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 20

The correct answer is $\frac{41}{81}$. An expression of the form $\sqrt[n]{a^m}$, where m and n are integers greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{m}{n}}$. Therefore, the expression on the left-hand side of the given equation is equivalent to $p^{\frac{2}{3}}$; thus, $p^{\frac{2}{3}} = t^{\frac{9}{7}}$. If $t = p^{3n-1}$, where n is a constant, then substituting p^{3n-1} for t in the equation yields $p^{\frac{2}{3}} = p^{\frac{9}{7}(3n-1)}$. It's given that $p > 1$; therefore, $\frac{2}{3} = \frac{9}{7}(3n-1)$. Multiplying each side of this equation by 21 yields $14 = 27(3n-1)$. Distributing multiplication over subtraction yields $14 = 81n - 27$. Adding 27 to each side of this equation yields $41 = 81n$. Dividing each side of this equation by 81 yields $\frac{41}{81} = n$. Therefore, if $t = p^{3n-1}$, where n is a constant, the value of n is $\frac{41}{81}$. Note that $41/81$, .5061, .5062, and 0.506 are examples of ways to enter a correct answer.

QUESTION 21

The correct answer is 302.4. It's given that the number b is 320% greater than 160. It follows that b is equal to 160 plus 320% of 160, which can be written as $b = 160 + 3.2(160)$, or $b = 672$. It's also given that the number a is 55% less than the number b . It follows that a is equal to b minus 55% of b , which can be written as $a = b - 0.55b$, or $a = 0.45b$. Substituting 672 for b in this equation yields $a = 0.45(672)$, or $a = 302.4$.

QUESTION 22

Choice C is correct. The given equation is quadratic. The maximum value of a function defined by a quadratic equation can be displayed as a constant in the vertex form of the equation, $f(x) = a(x - h)^2 + k$, where the maximum value of the function is k , which occurs when $x = h$, and a is a constant. The given equation can be rewritten in this form by completing the square. To complete the square, the given equation can be rewritten as $f(x) = -6(x^2 - 10x) - 126$, which is equivalent to $f(x) = -6(x^2 - 10x + 25) - 126 + 6(25)$, or $f(x) = -6(x - 5)^2 + 24$. This equation is in vertex form, where $a = -6$, $h = 5$, and $k = 24$. Therefore, $f(x) = -6(x - 5)^2 + 24$ displays the maximum value, 24, of the function as a constant.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This form displays one of the zeros, not the maximum value, of the function as a constant. **Choice D** is incorrect. This form displays the zeros, not the maximum value, of the function as constants.

QUESTION 23

Choice A is correct. It's given that $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . It follows that $\angle WQX$ and $\angle YQZ$ are vertical angles, which means they are congruent to each other. The figure shows that $\angle W$ and $\angle Y$ both have a measure of a° , so they are congruent to each other. Therefore, $\triangle QWX$ and $\triangle QYZ$ are similar triangles, where $\overline{WQ}$ corresponds to $\overline{YQ}$ and $\overline{WX}$ corresponds to $\overline{YZ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{YQ} = \frac{WX}{WQ}$. It's given that $YQ = 21$, $WQ = 70$, and $WX = 60$. Substituting 21 for YQ , 70 for WQ , and 60 for WX in the equation $\frac{YZ}{YQ} = \frac{WX}{WQ}$ yields $\frac{YZ}{21} = \frac{60}{70}$, or $\frac{YZ}{21} = \frac{6}{7}$. Multiplying each side of this equation by 21 yields $YZ = 18$. Therefore, the length of $\overline{YZ}$ is 18.

Choice B is incorrect. This is the length of $\overline{QZ}$, not $\overline{YZ}$. **Choice C** is incorrect. This is the length of $\overline{XQ}$, not $\overline{YZ}$. **Choice D** is incorrect and may result from conceptual or calculation errors.

QUESTION 24

Choice B is correct. The given equation can be rewritten as $52(x + 4)(x^2 - 4x + 16)(x^2 - 9)(x^2 + 9) = 0$, which is equivalent to $52(x + 4)(x^2 - 4x + 16)(x - 3)(x + 3)(x^2 + 9) = 0$. Applying the zero product property in this equation yields $52 = 0$, $x + 4 = 0$, $x^2 - 4x + 16 = 0$, $x - 3 = 0$, $x + 3 = 0$, and $x^2 + 9 = 0$. The equation $52 = 0$ has no solution. Subtracting 4 from both sides of the equation $x + 4 = 0$ yields $x = -4$. Adding 3 to both sides of the equation $x - 3 = 0$ yields $x = 3$. Subtracting 3 from both sides of the equation $x + 3 = 0$ yields $x = -3$. The discriminant of a quadratic equation of the form $ax^2 + bx + c = 0$ is $b^2 - 4ac$. If the discriminant is positive, the equation has two distinct real solutions. If the discriminant is zero, the equation has one distinct real solution. If the discriminant is negative, the equation has no real solutions. In the equation $x^2 - 4x + 16 = 0$, $a = 1$, $b = -4$, and $c = 16$. Substituting these values in the equation for the discriminant, $b^2 - 4ac$, yields $(-4)^2 - 4(1)(16)$, or -48 . Since

the discriminant is negative, this equation has no real solutions. In the equation $x^2 + 9 = 0$, $a = 1$, $b = 0$, and $c = 9$. Substituting these values in the equation for the discriminant, $b^2 - 4ac$, yields $0^2 - 4(1)(9)$, or -36 . Since the discriminant is negative, this equation has no real solutions. Therefore, the distinct real solutions to the given equation are $x = -4$, $x = 3$, and $x = -3$. Thus, the given equation has exactly three distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 25

Choice B is correct. The mean of a data set is the sum of the values divided by the number of values. It's given that a new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. The histogram shows that all the values in data set A are greater than 18. Since the additional number of points, 18, is less than any of the values in the original data set, the mean number of points per player for data set B is less than the mean number of points per player for data set A. The median of a data set with 50 values is between the 25th and 26th values when the values are listed in ascending order. The median of a data set with 51 values is the 26th value when the values are listed in ascending order. Therefore, the median of data set B is the 26th value in data set B, which is the 25th value in data set A. The histogram shows that the 25th and 26th values in data set A are both between 50 and 60 points. Therefore, the 25th and 26th values in data set A could be equal, in which case the median of data set A would be equal to both the 25th and 26th values in data set A. Then, the median number of points per player for data set A would be equal to the median number of points per player for data set B. Therefore, only statement II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 26

Choice D is correct. It's given that in triangle XYZ, the measure of angle X is 90° , so triangle XYZ is a right triangle. It's also given that point W lies on segment YZ, and segment WX is perpendicular to segment YZ. It follows that triangle WYX is a right triangle. Triangle WYX and triangle XYZ are right triangles with the same interior angle measure at point Y. It follows that triangle WYX and triangle XYZ are similar, where angles W, Y, and X in triangle WYX correspond to angles X, Y, and Z, respectively, in triangle XYZ. Since corresponding angles in similar triangles are congruent, their tangents have equal value. Thus, the tangent of angle Z is equal to the tangent of angle X in triangle WYX. The opposite side of angle X in triangle WYX is segment WY, and the adjacent side is segment WX. It's given that the length of segment WY is 572 and the length of segment WX is 429. It follows that the tangent of angle X in triangle WYX is $\frac{572}{429}$, or $\frac{4}{3}$. Therefore, the value of $\tan Z$ is $\frac{4}{3}$.

Choice A is incorrect. This is the value of $\cos Z$, not $\tan Z$. *Choice B* is incorrect. This is the value of $\tan Y$, not $\tan Z$. *Choice C* is incorrect. This is the value of $\sin Z$, not $\tan Z$.

QUESTION 27

The correct answer is 157.8. It's given that an area of 46.00 square nautical miles is equivalent to k square kilometers. Since 1 nautical mile is equal to 1.852 kilometers, it follows that an area of 1 square nautical mile is equivalent to $(1.852)^2$, or 3.429904, square kilometers. Multiplying 46.00 by 3.429904 yields 157.775584 square kilometers. Rounding this value to the nearest tenth yields 157.8. Therefore, the value of k is 157.8.